

## Unit 2. Multiple Regression I: Matrix Theory

# 1 The Multiple Regression Model

FREES §3.2 · ISLR §3.2.1 · CLM §1.

## 1.1 Model and matrix notation

**Definition 2.1** (Multiple Linear Regression Model). Given  $n$  observations, each consisting of a dependent variable  $Y_i$  and  $k$  explanatory variables  $X_{i1}, \dots, X_{ik}$ , the **multiple linear regression model** assumes

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \varepsilon_i, \quad i = 1, \dots, n, \quad (1)$$

equivalently, in matrix form,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (2)$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}_{n \times 1}, \quad \mathbf{X} = \begin{pmatrix} 1 & X_{11} & \cdots & X_{1k} \\ 1 & X_{21} & \cdots & X_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{nk} \end{pmatrix}_{n \times (k+1)}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

$\mathbf{X}$  is the **design matrix**; its first column is  $\mathbf{1}$ , carrying the intercept. Writing  $\mathbf{x}'_i$  for the  $i$ -th row of  $\mathbf{X}$  gives  $Y_i = \mathbf{x}'_i \boldsymbol{\beta} + \varepsilon_i$ .

**Remark** (Counts).  $k$  is the number of explanatory variables, excluding the intercept;  $k + 1$  is the number of parameters, i.e. the number of columns of  $\mathbf{X}$ ;  $n - k - 1$  is the residual degrees of freedom. ISLR writes  $p$  for  $k$  and  $p + 1$  for  $k + 1$ ; FREES writes  $k$ , as here.

## 1.2 Classical assumptions

**Definition 2.2** (Classical Assumptions). The following are imposed on model (2):

- (A1) **Linearity:**  $E[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$ .
- (A2) **Zero conditional mean:**  $E[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$ .
- (A3) **Spherical errors:**  $\text{var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$ ; that is,  $\text{var}(\varepsilon_i) = \sigma^2$  for every  $i$  (homoscedasticity) and  $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$  for  $i \neq j$  (no serial correlation).
- (A4) **Full rank:**  $\text{rank}(\mathbf{X}) = k + 1$ , requiring  $n \geq k + 1$ ; inference requires  $n > k + 1$  so that  $\text{MSE} = \text{SSE}/(n - k - 1)$  is defined.
- (A5) **Normality:**  $\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ .

A1–A4 are the **Gauss–Markov assumptions**; A5 is needed only for exact finite-sample distributions.

**Remark.**  $\mathbf{X}$  is treated as fixed and all expectations, variances and distributions are conditional on it, with the conditioning suppressed.

**Remark** (A1 and A2 are equivalent). Taking conditional expectations in  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  gives  $\mathbf{E}[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta} + \mathbf{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}]$ , so each holds if and only if the other does. Both are stated because A1 matches how the model is written and A2 how residuals are handled.

**Remark** (Which assumption carries which result).

| Result                                                                        | Requires | Fails under                             |
|-------------------------------------------------------------------------------|----------|-----------------------------------------|
| $\hat{\boldsymbol{\beta}}$ exists, is unique                                  | A4       | perfect collinearity                    |
| $\hat{\boldsymbol{\beta}}$ unbiased                                           | A1, A2   | omitted variable, wrong functional form |
| $\text{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ | + A3     | heteroscedasticity, autocorrelation     |
| $\hat{\boldsymbol{\beta}}$ is BLUE                                            | A1–A4    | any of the above                        |
| $t$ , $F$ exact                                                               | + A5     | heavy tails, small $n$                  |

Normality is required for none of the first four rows.

## 2 Least Squares in Matrix Form

FREES §3.1 · ISLR §3.2.1 · CLM §2.1–2.3 · MC §4, §6.

## 2.1 Normal equations

**Definition 2.3** (Ordinary Least Squares). The **OLS estimator**  $\hat{\beta}$  minimises

$$Q(\beta) = \sum_{i=1}^n (Y_i - \mathbf{x}_i' \beta)^2 = (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|^2. \quad (3)$$

**Lemma 2.4** (Matrix differentiation). For  $\beta \in \mathbb{R}^p$ , constant  $\mathbf{a} \in \mathbb{R}^p$  and constant  $\mathbf{A} \in \mathbb{R}^{p \times p}$ , write  $Df(\beta) := \partial f / \partial \beta'$  (a row vector). Then

- (i)  $D(\mathbf{a}'\beta) = D(\beta'\mathbf{a}) = \mathbf{a}'$ ;
- (ii)  $D(\beta'\mathbf{A}\beta) = \beta'(\mathbf{A} + \mathbf{A}')$ , which is  $2\beta'\mathbf{A}$  when  $\mathbf{A}' = \mathbf{A}$ .

**Proof.** Apply the first identification theorem,  $df(\beta) = \mathbf{c}(\beta)'d\beta \iff Df(\beta) = \mathbf{c}(\beta)'$ .

(i)  $d(\mathbf{a}'\beta) = \mathbf{a}'d\beta$  since  $d\mathbf{a} = 0$ ; and  $\beta'\mathbf{a} = \mathbf{a}'\beta$ .

(ii) With  $d\mathbf{A} = 0$ , the product rule and  $(d\beta)'\mathbf{A}\beta = \beta'\mathbf{A}'d\beta$  give

$$d(\beta'\mathbf{A}\beta) = (d\beta)'\mathbf{A}\beta + \beta'\mathbf{A}d\beta = \beta'(\mathbf{A} + \mathbf{A}')d\beta. \quad \square$$

**Theorem 2.5** (OLS estimator). Under A4,

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}. \quad (4)$$

**Proof. Step 1.** Expand, using that the scalar  $\mathbf{Y}'\mathbf{X}\beta$  equals its transpose:

$$Q(\beta) = \mathbf{Y}'\mathbf{Y} - \mathbf{Y}'\mathbf{X}\beta - \beta'\mathbf{X}'\mathbf{Y} + \beta'\mathbf{X}'\mathbf{X}\beta = \mathbf{Y}'\mathbf{Y} - 2\beta'\mathbf{X}'\mathbf{Y} + \beta'\mathbf{X}'\mathbf{X}\beta. \quad (5)$$

**Step 2.** By Lemma 2.4 with  $\mathbf{a} = \mathbf{X}'\mathbf{Y}$ ,  $\mathbf{A} = \mathbf{X}'\mathbf{X}$  symmetric,  $DQ(\beta) = -2\mathbf{Y}'\mathbf{X} + 2\beta'\mathbf{X}'\mathbf{X}$ . Setting  $DQ = \mathbf{0}'$  and transposing gives the **normal equations**

$$\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{Y}. \quad (6)$$

**Step 3.**  $\text{rank}(\mathbf{X}'\mathbf{X}) = \text{rank}(\mathbf{X})$ : if  $\mathbf{X}\mathbf{v} = \mathbf{0}$  then  $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$ ; conversely  $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$  gives  $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 = 0$ , hence  $\mathbf{X}\mathbf{v} = \mathbf{0}$ . The null spaces coincide, so the ranks do. By A4 that rank is  $k + 1$ , so  $\mathbf{X}'\mathbf{X}$  is invertible and (6) gives (4).

**Step 4.** The Hessian is  $2\mathbf{X}'\mathbf{X}$ , and  $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 > 0$  for  $\mathbf{v} \neq \mathbf{0}$  by A4: positive definite, so  $\hat{\beta}$  is a global minimum.  $\square$

**Remark.** Only Step 3 uses A4. Without it  $\mathbf{X}'\mathbf{X}$  is singular and (6) has infinitely many solutions; R reports NA for the redundant coefficients rather than raising an error. Near-singularity leaves  $\mathbf{X}'\mathbf{X}$  invertible but ill-conditioned, so  $\hat{\boldsymbol{\beta}}$  exists and is unstable.

## 2.2 Orthogonal projection

**Theorem 2.6** (OLS as orthogonal projection). The fitted vector  $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$  is the orthogonal projection of  $\mathbf{Y}$  onto  $\mathcal{C}(\mathbf{X})$ , and  $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ .

**Proof.** Minimising  $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$  is finding the point of  $\mathcal{C}(\mathbf{X})$  closest to  $\mathbf{Y}$ .

**Step 1.**  $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$  minimises  $\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2$  over  $\tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$  iff  $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$ .

( $\Leftarrow$ ) For  $\tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ ,  $\hat{\mathbf{Y}} - \tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ , so  $(\mathbf{Y} - \hat{\mathbf{Y}}) \perp (\hat{\mathbf{Y}} - \tilde{\mathbf{Y}})$  and Pythagoras gives

$$\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 + \|\hat{\mathbf{Y}} - \tilde{\mathbf{Y}}\|^2 \geq \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2,$$

with equality iff  $\tilde{\mathbf{Y}} = \hat{\mathbf{Y}}$ .

( $\Rightarrow$ ) If some  $\mathbf{v} \in \mathcal{C}(\mathbf{X})$  with  $\|\mathbf{v}\| = 1$  had  $\mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) \neq 0$ , then for  $\tilde{\mathbf{Y}} = \hat{\mathbf{Y}} + t\mathbf{v}$ ,

$$\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 - 2t\mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) + t^2,$$

strictly smaller at  $t^* = \mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}})$ , contradicting optimality.

**Step 2.** The residual is orthogonal to  $\mathcal{C}(\mathbf{X})$  iff orthogonal to each column of  $\mathbf{X}$ , i.e.  $\mathbf{X}'(\mathbf{Y} - \hat{\mathbf{Y}}) = \mathbf{0}$ ; with  $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$  this is (6).

**Step 3.** A4 makes  $\mathbf{X}'\mathbf{X}$  invertible. □

## 3 Hat Matrix, Residuals, Leverage

FREES §3.2 · ISLR §3.3.3 · CLM §2.4–2.7.

### 3.1 Hat matrix and residual maker

**Definition 2.7** (Hat matrix, residual maker).

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \quad \mathbf{M} = \mathbf{I}_n - \mathbf{H}, \quad (7)$$

so that  $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$  and  $\mathbf{e} = \mathbf{M}\mathbf{Y}$ .

**Remark** (Errors versus residuals). The **error**  $\varepsilon_i = Y_i - \mathbf{x}_i'\boldsymbol{\beta}$  is unobservable; the **residual**

$e_i = Y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}}$  is observable. Since  $\mathbf{M}\mathbf{X} = \mathbf{X} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{0}$ ,

$$\mathbf{e} = \mathbf{M}\mathbf{Y} = \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{M}\boldsymbol{\varepsilon}, \quad (8)$$

a rank-deficient linear map of  $\boldsymbol{\varepsilon}$ ; this is the source of the  $n - k - 1$  degrees of freedom.

**Definition 2.8** (Idempotent matrix). A square matrix  $\mathbf{A}$  is **idempotent** if  $\mathbf{A}^2 = \mathbf{A}$ .

**Lemma 2.9** (Trace equals rank). If  $\mathbf{A}$  is  $n \times n$  and idempotent, then  $\text{tr}(\mathbf{A}) = \text{rank}(\mathbf{A})$ .

**Proof. Step 1.** If  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ ,  $\mathbf{v} \neq \mathbf{0}$ , then  $\lambda\mathbf{v} = \mathbf{A}^2\mathbf{v} = \lambda^2\mathbf{v}$ , so  $\lambda \in \{0, 1\}$ .

**Step 2.**  $\mathbb{R}^n = \mathcal{N}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})$ : any  $\mathbf{v}$  splits as  $(\mathbf{v} - \mathbf{A}\mathbf{v}) + \mathbf{A}\mathbf{v}$ , with  $\mathbf{A}(\mathbf{v} - \mathbf{A}\mathbf{v}) = \mathbf{0}$ ; and if  $\mathbf{v} = \mathbf{A}\mathbf{w} \in \mathcal{C}(\mathbf{A}) \cap \mathcal{N}(\mathbf{A})$  then  $\mathbf{v} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{v} = \mathbf{0}$ .

**Step 3.** Every  $\mathbf{u} \in \mathcal{C}(\mathbf{A})$  has  $\mathbf{A}\mathbf{u} = \mathbf{u}$ . A basis of  $\mathcal{N}(\mathbf{A})$  (eigenvalue 0) together with a basis of  $\mathcal{C}(\mathbf{A})$  (eigenvalue 1) is an eigenvector basis, so  $\mathbf{A} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}^{-1}$  with  $\boldsymbol{\Lambda}$  diagonal, entries in  $\{0, 1\}$ .

**Step 4.** With  $r = \text{rank}(\mathbf{A})$ ,  $\boldsymbol{\Lambda}$  has  $r$  ones, so  $\text{tr}(\mathbf{A}) = \text{tr}(\boldsymbol{\Lambda}) = r$ .  $\square$

**Theorem 2.10** (Properties of  $\mathbf{H}$  and  $\mathbf{M}$ ).

- (i)  $\mathbf{H}$  and  $\mathbf{M}$  are symmetric and idempotent.
- (ii)  $\mathbf{HX} = \mathbf{X}$  and  $\mathbf{MX} = \mathbf{0}$ .
- (iii)  $\mathbf{HM} = \mathbf{MH} = \mathbf{0}$ .
- (iv)  $\text{tr}(\mathbf{H}) = k + 1$  and  $\text{tr}(\mathbf{M}) = n - k - 1$ .

**Proof.** (i) Since  $(\mathbf{X}'\mathbf{X})^{-1}$  is symmetric,

$$\mathbf{H}' = \mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}]'\mathbf{X}' = \mathbf{H}.$$

Moreover,

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \underbrace{\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}}_{\mathbf{I}_{k+1}} \mathbf{X}' = \mathbf{H}.$$

Then  $\mathbf{M}' = \mathbf{I} - \mathbf{H}' = \mathbf{M}$  and  $\mathbf{M}^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{M}$ .

(ii)  $\mathbf{HX} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{X}$ , hence  $\mathbf{MX} = \mathbf{0}$ .

(iii)  $\mathbf{HM} = \mathbf{H} - \mathbf{H}^2 = \mathbf{0}$ ; likewise  $\mathbf{MH} = \mathbf{0}$ .

(iv) By Lemma 2.9,  $\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H})$ . Now  $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ : every column of  $\mathbf{H}$  is  $\mathbf{X}$  times a vector, and  $\mathbf{H}\mathbf{X} = \mathbf{X}$  from (ii) puts every column of  $\mathbf{X}$  in  $\mathcal{C}(\mathbf{H})$ . By A4,  $\dim \mathcal{C}(\mathbf{X}) = k + 1$ , so  $\text{tr}(\mathbf{H}) = k + 1$  and  $\text{tr}(\mathbf{M}) = n - k - 1$ .  $\square$

**Remark.**  $\mathbf{H}$  is symmetric, idempotent, with  $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ : these three properties characterise the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$ , so Theorem 2.6 and Definition 2.7 describe one object.  $\mathbf{M}$  projects onto  $\mathcal{C}(\mathbf{X})^\perp$ , of dimension  $n - k - 1$ .

## 3.2 Residual identities

**Theorem 2.11** (Implications of the normal equations). The OLS residuals satisfy

- (i)  $\mathbf{X}'\mathbf{e} = \mathbf{0}$ ;
- (ii)  $\sum_i e_i = 0$ , provided the model contains an intercept;
- (iii)  $\hat{\mathbf{Y}}'\mathbf{e} = 0$ .

**Proof.** (i) Rearranging (6),  $\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$ . (ii) The first column of  $\mathbf{X}$  is  $\mathbf{1}$ , so the first component of  $\mathbf{X}'\mathbf{e} = \mathbf{0}$  is  $\mathbf{1}'\mathbf{e} = \sum_i e_i = 0$ . (iii)  $\hat{\mathbf{Y}}'\mathbf{e} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{e} = 0$  by (i).  $\square$

**Remark.** These hold for every data set, whatever the fit: they follow from the choice of  $\hat{\beta}$ , not from the model being correct. A residual mean of zero is therefore not evidence of anything, and the identities place no constraint on structure **within** the residuals, which is where a misspecified model shows.

### 3.3 Leverage

**Definition 2.12** (Leverage). The **leverage** of observation  $i$  is  $h_{ii}$ , the  $i$ -th diagonal entry of  $\mathbf{H}$ .

**Theorem 2.13** (Properties of leverage).

- (i)  $0 \leq h_{ii} \leq 1$ ;
- (ii)  $\sum_{i=1}^n h_{ii} = k + 1$ , so the mean leverage is  $(k + 1)/n$ .

**Proof.** (i)  $\mathbf{H} = \mathbf{H}'\mathbf{H}$ , so comparing  $(i, i)$  entries with  $h_{ji} = h_{ij}$ ,

$$h_{ii} = \sum_{j=1}^n h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2.$$

The left equality gives  $h_{ii} \geq 0$ ; dropping the tail gives  $h_{ii}(1 - h_{ii}) \geq 0$ , hence  $h_{ii} \leq 1$ . **(ii)**  
 $\sum_i h_{ii} = \text{tr}(\mathbf{H}) = k + 1$  by Theorem 2.10(iv).  $\square$

**Remark.**  $h_{ii}$  depends on  $\mathbf{X}$  alone, not on  $\mathbf{Y}$ : it measures how extreme the predictor values are. The conventional flag is  $h_{ii} > 2(k + 1)/n$ , twice the mean.

End of period 1 of 3.

## 4 Finite-Sample Properties

FREES §3.2 · ISLR §3.2.1 · CLM §3. Only A1–A4 are used.

**Definition 2.14** (Sums of squares and mean squares). For the fitted model  $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ ,

$$\text{SST} = \sum_i (Y_i - \bar{Y})^2, \quad \text{SSR} = \sum_i (\hat{Y}_i - \bar{Y})^2, \quad \text{SSE} = \sum_i (Y_i - \hat{Y}_i)^2 = \mathbf{e}'\mathbf{e},$$

$$\text{MSE} = s^2 = \frac{\text{SSE}}{n - k - 1}, \quad \text{MSR} = \frac{\text{SSR}}{k}.$$

**Remark** (Notation). FREES writes SSE and SST; ISLR writes RSS and TSS for the same two quantities. Both spellings occur in the SOA sample questions.

**Lemma 2.15** (Variance under a linear map). If  $\text{var}(\mathbf{z})$  is defined,  $\mathbf{A}$  is  $m \times n$  constant and  $\mathbf{b} \in \mathbb{R}^m$  constant, then  $\text{var}(\mathbf{Az} + \mathbf{b}) = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{A}'$ . In particular  $\text{var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$  under A1–A3.

**Proof.** With  $\boldsymbol{\mu} = \mathbb{E}[\mathbf{z}]$ ,  $\mathbb{E}[\mathbf{Az} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$ , so

$$\text{var}(\mathbf{Az} + \mathbf{b}) = \mathbb{E}[\mathbf{A}(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})' \mathbf{A}'] = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{A}'.$$

Applying this to  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  gives  $\text{var}(\mathbf{Y}) = \text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$  by A3. □

## 4.1 Unbiasedness and variance

**Theorem 2.16** (Unbiasedness). Under A1–A2,  $E[\hat{\beta}] = \beta$ .

**Proof.** Substituting  $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$  into  $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ ,

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon, \quad (9)$$

and  $E[\varepsilon] = \mathbf{0}$  by A2. □

**Theorem 2.17** (Variance–covariance matrix). Under A1–A4,  $\text{var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ .

**Proof.** By (9) and Lemma 2.15 with  $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ ,

$$\text{var}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.$$

The  $j$ -th diagonal entry is  $\text{var}(\hat{\beta}_j) = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$ . □

**Remark.** Three consequences: standard errors scale with  $\sigma$ ; well-spread predictors make  $\mathbf{X}'\mathbf{X}$  large and  $(\mathbf{X}'\mathbf{X})^{-1}$  small; and correlated predictors make  $\mathbf{X}'\mathbf{X}$  near-singular and inflate  $(\mathbf{X}'\mathbf{X})^{-1}$ .

## 4.2 The Gauss–Markov theorem

**Theorem 2.18** (Gauss–Markov). Under A1–A4,  $\hat{\boldsymbol{\beta}}$  is the best linear unbiased estimator of  $\boldsymbol{\beta}$ : for any linear unbiased  $\tilde{\boldsymbol{\beta}} = \mathbf{C}\mathbf{Y}$  with  $\mathbf{C}$  constant,  $\text{var}(\tilde{\boldsymbol{\beta}}) - \text{var}(\hat{\boldsymbol{\beta}})$  is positive semi-definite, so  $\text{var}(\tilde{\beta}_j) \geq \text{var}(\hat{\beta}_j)$  for every  $j$ .

**Proof. Step 1.**  $E[\tilde{\boldsymbol{\beta}}] = \mathbf{C}\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$  for all  $\boldsymbol{\beta}$  forces  $\mathbf{C}\mathbf{X} = \mathbf{I}_{k+1}$ .

**Step 2.** Write  $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$ ; then  $\mathbf{D}\mathbf{X} = \mathbf{C}\mathbf{X} - \mathbf{I}_{k+1} = \mathbf{0}$ .

**Step 3.** By Lemma 2.15,  $\text{var}(\tilde{\boldsymbol{\beta}}) = \sigma^2\mathbf{C}\mathbf{C}'$ , which expands to

$$\sigma^2\mathbf{C}\mathbf{C}' = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \quad (\text{A})$$

$$+ \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}' \quad (\text{B})$$

$$+ \sigma^2\mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \quad (\text{C})$$

$$+ \sigma^2\mathbf{D}\mathbf{D}'. \quad (\text{D})$$

(A) is  $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ ; (C) vanishes since  $\mathbf{DX} = \mathbf{0}$ ; (B) is its transpose and also vanishes. Hence  $\text{var}(\tilde{\boldsymbol{\beta}}) = \text{var}(\hat{\boldsymbol{\beta}}) + \sigma^2\mathbf{DD}'$ , and  $\mathbf{v}'\mathbf{DD}'\mathbf{v} = \|\mathbf{D}'\mathbf{v}\|^2 \geq 0$ . Equality forces  $\mathbf{D} = \mathbf{0}$ .  $\square$

**Remark.** The theorem ranks OLS only within the class of estimators linear in  $\mathbf{Y}$  and unbiased. Dropping unbiasedness admits estimators with smaller mean squared error, ridge among them; no normality is used, and nothing is claimed about prediction error on new data.

## 4.3 Estimating $\sigma^2$

**Lemma 2.19** (Expectation of a quadratic form). If  $\mathbb{E}[\mathbf{z}] = \boldsymbol{\mu}$ ,  $\text{var}(\mathbf{z}) = \boldsymbol{\Sigma}$  and  $\mathbf{A}$  is constant  $n \times n$ , then  $\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$ .

**Proof.**  $\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \sum_{i,j} a_{ij} \mathbb{E}[z_i z_j]$  and  $\mathbb{E}[z_i z_j] = \Sigma_{ij} + \mu_i \mu_j$ . The first part is  $\sum_{i,j} a_{ij} \Sigma_{ij} = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$  by symmetry of  $\boldsymbol{\Sigma}$ ; the second is  $\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$ .  $\square$

**Theorem 2.20** (Unbiasedness of MSE). Under A1–A4,  $\mathbb{E}[\text{MSE}] = \sigma^2$ .

**Proof.** By (8) and symmetric idempotency of  $\mathbf{M}$ ,  $\text{SSE} = \boldsymbol{\varepsilon}'\mathbf{M}'\mathbf{M}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$ . Lemma 2.19 with  $\boldsymbol{\mu} = \mathbf{0}$ ,  $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$  gives  $\mathbf{E}[\text{SSE}] = \sigma^2 \text{tr}(\mathbf{M}) = \sigma^2(n - k - 1)$  by Theorem 2.10(iv).  $\square$

**Remark.** The divisor is  $n - k - 1$  because  $\text{tr}(\mathbf{M}) = n - \text{tr}(\mathbf{H}) = n - (k + 1)$ : the residual vector lies in the  $(n - k - 1)$ -dimensional space  $\mathcal{C}(\mathbf{X})^\perp$ . Dividing by  $n$  instead gives an estimator biased downward by the factor  $(n - k - 1)/n$ .

**Theorem 2.21** (Variance of the residuals). Under A1–A4,  $\text{var}(\mathbf{e}) = \sigma^2\mathbf{M}$ , so  $\text{var}(e_i) = \sigma^2(1 - h_{ii})$ .

**Proof.**  $\text{var}(\mathbf{e}) = \mathbf{M}(\sigma^2\mathbf{I})\mathbf{M}' = \sigma^2\mathbf{M}^2 = \sigma^2\mathbf{M}$ ; the  $(i, i)$  entry of  $\mathbf{I} - \mathbf{H}$  is  $1 - h_{ii}$ .  $\square$

**Remark.** Residuals are heteroscedastic by construction even when the errors are not, so residual plots use the **standardised** residual  $e_i/(s\sqrt{1 - h_{ii}})$ , which is **rstandard** in R.

## 5 Exact Inference Under Normality

CLM §4–5; FREES §3.4 uses these distributions. A5 is now in force.

## 5.1 Multivariate normal tools

**Theorem 2.22** (Closure under linear maps). If  $\mathbf{z} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ ,  $\mathbf{A}$  is  $m \times n$  constant and  $\mathbf{b} \in \mathbb{R}^m$  constant, then  $\mathbf{Az} + \mathbf{b} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$ .

**Proof.** With  $M_{\mathbf{z}}(\mathbf{t}) = \exp(\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t})$ ,

$$M_{\mathbf{Az}+\mathbf{b}}(\mathbf{t}) = e^{\mathbf{t}'\mathbf{b}} M_{\mathbf{z}}(\mathbf{A}'\mathbf{t}) = \exp(\mathbf{t}'(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2}\mathbf{t}'\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'\mathbf{t}),$$

the mgf of  $N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$ ; the mgf is unique. □

**Theorem 2.23** (Uncorrelated  $\Leftrightarrow$  independent, jointly normal). If  $(\mathbf{U}', \mathbf{V}')$  is jointly normal with cross-covariance  $\boldsymbol{\Sigma}_{UV}$ , then  $\boldsymbol{\Sigma}_{UV} = \mathbf{0}$  iff  $\mathbf{U}$  and  $\mathbf{V}$  are independent.

**Proof.** ( $\Leftarrow$ ) holds for any distribution. ( $\Rightarrow$ ) With  $\boldsymbol{\Sigma}_{UV} = \mathbf{0}$  the covariance is block diagonal, so

$$M(\mathbf{s}, \mathbf{t}) = \exp(\mathbf{s}'\boldsymbol{\mu}_U + \frac{1}{2}\mathbf{s}'\boldsymbol{\Sigma}_{UU}\mathbf{s}) \exp(\mathbf{t}'\boldsymbol{\mu}_V + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}_{VV}\mathbf{t}) = M_{\mathbf{U}}(\mathbf{s})M_{\mathbf{V}}(\mathbf{t}). \quad \square$$

**Remark.** For general distributions uncorrelatedness does not imply independence; the equivalence is special to the multivariate normal, and is what makes exact  $t$  and  $F$  inference possible.

**Theorem 2.24** (Chi-square quadratic forms). If  $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$  and  $\mathbf{A}$  is symmetric idempotent, then  $\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi_r^2$  with  $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$ .

**Proof.** Write  $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$  with  $\mathbf{P}$  orthogonal; by Lemma 2.9 the eigenvalues are  $r$  ones and  $n - r$  zeros. With  $\mathbf{w} = \mathbf{P}'\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$  by Theorem 2.22,  $\mathbf{z}'\mathbf{A}\mathbf{z} = \mathbf{w}'\mathbf{\Lambda}\mathbf{w} = \sum_{i=1}^r w_i^2 \sim \chi_r^2$ .  $\square$

**Theorem 2.25** (Cochran). If  $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$  and  $\mathbf{A}_1, \mathbf{A}_2$  are symmetric idempotent with  $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$ , then  $\mathbf{z}'\mathbf{A}_1\mathbf{z}$  and  $\mathbf{z}'\mathbf{A}_2\mathbf{z}$  are independent.

**Proof.**  $(\mathbf{U}', \mathbf{V}')' = (\mathbf{A}'_1, \mathbf{A}'_2)'\mathbf{z}$  is one linear map of one normal vector, hence jointly normal by Theorem 2.22; individual normality would not suffice. Its cross-covariance is  $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$ , so  $\mathbf{U}$  and  $\mathbf{V}$  are independent by Theorem 2.23, and  $\mathbf{z}'\mathbf{A}_i\mathbf{z} = \|\mathbf{A}_i\mathbf{z}\|^2$  is a function of one of them alone.  $\square$

## 5.2 Sampling distributions

**Theorem 2.26** (Normality of  $\hat{\beta}$ ). Under A1–A5,  $\hat{\beta} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$ .

**Proof.** By (9),  $\hat{\beta}$  is an affine map of  $\varepsilon \sim N(\mathbf{0}, \sigma^2\mathbf{I})$ ; apply Theorem 2.22 with  $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ . □

**Theorem 2.27** (Distribution of SSE; independence from  $\hat{\beta}$ ). Under A1–A5,  $\text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ , and SSE is independent of  $\hat{\beta}$ .

**Proof. Step 1.** With  $\mathbf{z} = \varepsilon/\sigma \sim N(\mathbf{0}, \mathbf{I}_n)$ ,  $\text{SSE}/\sigma^2 = \mathbf{z}'\mathbf{M}\mathbf{z}$ .

**Step 2.**  $\mathbf{M}$  is symmetric idempotent with  $\text{tr}(\mathbf{M}) = n - k - 1$ , so Theorem 2.24 applies.

**Step 3.** The stacked vector

$$\begin{pmatrix} \hat{\beta} - \beta \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ \mathbf{M} \end{pmatrix} \varepsilon$$

is jointly normal, with cross-covariance  $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0}$  since  $\mathbf{X}'\mathbf{M} = (\mathbf{M}\mathbf{X})' = \mathbf{0}$ . Apply Theorem 2.23. □

**Remark.** A  $t$  statistic is a standard normal over  $\sqrt{\chi_\nu^2/\nu}$  **with numerator and denominator independent**. Here the numerator comes from  $\hat{\beta}_j$  and the denominator from SSE; Step 3 supplies the independence, without which the ratio would not be  $t$ .

**Theorem 2.28** (OLS equals MLE under normality). Under A1–A5,  $\hat{\boldsymbol{\beta}}$  is the maximum likelihood estimator of  $\boldsymbol{\beta}$ , and  $\hat{\sigma}_{\text{MLE}}^2 = \text{SSE}/n$ , biased downward by the factor  $(n - k - 1)/n$ .

**Proof.**  $\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ , so

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})' (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}). \quad (10)$$

Only the last term involves  $\boldsymbol{\beta}$ , with negative coefficient  $-1/(2\sigma^2)$ , so maximising over  $\boldsymbol{\beta}$  minimises  $Q(\boldsymbol{\beta})$ . Substituting  $\hat{\boldsymbol{\beta}}$ ,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\text{SSE}}{2(\sigma^2)^2} = 0 \implies \hat{\sigma}_{\text{MLE}}^2 = \frac{\text{SSE}}{n},$$

and  $E[\text{SSE}] = (n - k - 1)\sigma^2$  by Theorem 2.20. □

**Remark.** Since OLS is maximum likelihood here, any criterion built from the maximised likelihood, AIC and BIC among them, can be computed from (10) with  $\hat{\sigma}_{\text{MLE}}^2$  substituted, and needs no further distribution theory.

End of period 2 of 3.

## 6 Goodness of Fit

FREES §3.3 · ISLR §3.2.2 · CLM §6.

### 6.1 Variance decomposition

**Definition 2.29** (Centering matrix).  $\mathbf{M}_0 = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$ , so that  $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1}$ .

**Remark.**  $\mathbf{M}_0$  is the residual maker for the intercept-only model,  $\mathbf{M}$  that for the full design:  $\mathbf{M}_0$  centers,  $\mathbf{M}$  regresses out all of  $\mathbf{X}$ . Both are symmetric idempotent, with  $\text{tr}(\mathbf{M}_0) = n - 1$  and  $\text{tr}(\mathbf{M}) = n - k - 1$ , projecting onto  $\mathbf{1}^\perp$  and  $\mathcal{C}(\mathbf{X})^\perp$  respectively. With an intercept,  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , so  $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0\mathbf{M} = \mathbf{M}$ .

**Theorem 2.30** (Variance decomposition). In a model with an intercept,

$$\boxed{\text{SST} = \text{SSR} + \text{SSE}}, \quad (11)$$

with  $\text{SST} = \mathbf{Y}'\mathbf{M}_0\mathbf{Y}$ ,  $\text{SSR} = \hat{\mathbf{Y}}'\mathbf{M}_0\hat{\mathbf{Y}}$  and  $\text{SSE} = \mathbf{e}'\mathbf{e}$ .

**Proof. Step 1.** For any  $\mathbf{v}$ , using  $\mathbf{M}_0'\mathbf{M}_0 = \mathbf{M}_0$ ,  $\mathbf{v}'\mathbf{M}_0\mathbf{v} = \|\mathbf{v} - \bar{v}\mathbf{1}\|^2 = \sum_i (v_i - \bar{v})^2$ . Take  $\mathbf{v} = \mathbf{Y}$  and  $\mathbf{v} = \hat{\mathbf{Y}}$ ; the latter needs  $\bar{\hat{Y}} = \bar{Y}$ , which follows from  $\mathbf{1}'\hat{\mathbf{Y}} = \mathbf{1}'(\mathbf{Y} - \mathbf{e}) = \mathbf{1}'\mathbf{Y}$  by Theorem 2.11(ii).

**Step 2.** Since  $\bar{e} = 0$ ,

$$\mathbf{M}_0\mathbf{Y} = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{M}_0\mathbf{e} = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e}. \quad (12)$$

**Step 3.** Expanding,  $\text{SST} = \text{SSR} + 2\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} + \text{SSE}$ .

**Step 4.**  $\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} = \mathbf{e}'\hat{\mathbf{Y}} - \bar{Y}\mathbf{e}'\mathbf{1} = 0$  by Theorem 2.11(ii),(iii). □

**Remark.** Equation (12) is orthogonal, so (11) is Pythagoras. Without an intercept  $\bar{e} \neq 0$ , the cross term survives and (11) is false.

## 6.2 $R^2$ and adjusted $R^2$

**Definition 2.31** (Coefficient of determination).

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}, \quad \bar{R}^2 = 1 - \frac{\text{SSE}/(n - k - 1)}{\text{SST}/(n - 1)} = 1 - \frac{n - 1}{n - k - 1}(1 - R^2). \quad (13)$$

$\bar{R}^2$  may be negative.

**Proposition 2.32** (Bounds). In a model with an intercept,  $0 \leq R^2 \leq 1$ .

**Proof.**  $\text{SSR} \geq 0$  and  $\text{SST} > 0$  give the lower bound;  $\text{SST} = \text{SSR} + \text{SSE}$  with  $\text{SSE} \geq 0$  gives the upper.  $\square$

**Theorem 2.33** (Monotonicity). Adding a regressor cannot decrease  $R^2$ .

**Proof.** With  $\mathcal{C}_k \subseteq \mathcal{C}_{k+1}$  the two column spaces, Theorem 2.6 makes  $\hat{\mathbf{Y}}_{k+1}$  the closest point of  $\mathcal{C}_{k+1}$  to  $\mathbf{Y}$ , and  $\hat{\mathbf{Y}}_k \in \mathcal{C}_k \subseteq \mathcal{C}_{k+1}$ , so  $\text{SSE}_{k+1} \leq \text{SSE}_k$ . SST does not depend on the regressors.  $\square$

**Remark.**  $R^2$  therefore cannot compare models of different size: adding pure noise raises it. The alternatives are penalised criteria ( $\bar{R}^2$ ,  $C_p$ , AIC, BIC) and out-of-sample error by cross-validation. Of these  $\bar{R}^2$  is the weakest: it rises whenever the new variable has  $|t| > 1$ .

## 6.3 ANOVA table and the overall $F$ test

| Source     | Sum of squares | df          | Mean square                           |
|------------|----------------|-------------|---------------------------------------|
| Regression | SSR            | $k$         | $\text{MSR} = \text{SSR}/k$           |
| Error      | SSE            | $n - k - 1$ | $\text{MSE} = \text{SSE}/(n - k - 1)$ |
| Total      | SST            | $n - 1$     |                                       |

**Lemma 2.34** (The regression projection). Let  $\mathbf{H}_0 = \frac{1}{n}\mathbf{1}\mathbf{1}'$  and suppose  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . Then  $\mathbf{H}\mathbf{H}_0 = \mathbf{H}_0\mathbf{H} = \mathbf{H}_0$ ;  $\mathbf{H} - \mathbf{H}_0$  is symmetric idempotent with  $\text{tr}(\mathbf{H} - \mathbf{H}_0) = k$ ; and  $(\mathbf{H} - \mathbf{H}_0)\mathbf{M} = \mathbf{0}$ .

**Proof.**  $\mathbf{H}$  projects onto  $\mathcal{C}(\mathbf{X}) \ni \mathbf{1}$ , so  $\mathbf{H}\mathbf{1} = \mathbf{1}$  and  $\mathbf{H}\mathbf{H}_0 = \frac{1}{n}(\mathbf{H}\mathbf{1})\mathbf{1}' = \mathbf{H}_0$ ; transposing gives  $\mathbf{H}_0\mathbf{H} = \mathbf{H}_0$ . Hence  $\mathbf{H} - \mathbf{H}_0$  is symmetric and

$$(\mathbf{H} - \mathbf{H}_0)^2 = \mathbf{H} - \mathbf{H}\mathbf{H}_0 - \mathbf{H}_0\mathbf{H} + \mathbf{H}_0 = \mathbf{H} - \mathbf{H}_0,$$

with  $\text{tr}(\mathbf{H} - \mathbf{H}_0) = \text{tr}(\mathbf{H}) - \text{tr}(\mathbf{H}_0) = (k + 1) - 1 = k$  by Theorem 2.10(iv). Finally  $(\mathbf{H} - \mathbf{H}_0)\mathbf{M} = \mathbf{H}\mathbf{M} - \mathbf{H}_0\mathbf{H}\mathbf{M} = \mathbf{0}$ , using  $\mathbf{H}\mathbf{M} = \mathbf{0}$  from Theorem 2.10(iii) and  $\mathbf{H}_0 = \mathbf{H}_0\mathbf{H}$ .  $\square$

**Theorem 2.35** (Overall  $F$  test). Under A1–A5, for  $H_0: \beta_1 = \cdots = \beta_k = 0$ ,

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{\text{SSR}/k}{\text{SSE}/(n - k - 1)} = \frac{R^2/k}{(1 - R^2)/(n - k - 1)} \sim F_{k, n-k-1}. \quad (14)$$

**Proof.** Under  $H_0$  the design collapses to the intercept, with projection  $\mathbf{H}_0 = \frac{1}{n}\mathbf{1}\mathbf{1}'$ . With  $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma$ , write  $\text{SSR}/\sigma^2 = \mathbf{z}'(\mathbf{H} - \mathbf{H}_0)\mathbf{z}$  and  $\text{SSE}/\sigma^2 = \mathbf{z}'\mathbf{M}\mathbf{z}$ . By Lemma 2.34 and Theorem 2.10, both  $\mathbf{H} - \mathbf{H}_0$  and  $\mathbf{M}$  are symmetric idempotent with traces  $k$  and  $n - k - 1$ , so Theorem 2.24 gives  $\chi_k^2$  and  $\chi_{n-k-1}^2$ ; and  $(\mathbf{H} - \mathbf{H}_0)\mathbf{M} = \mathbf{0}$ , so they are independent by Theorem 2.25. The ratio of independent chi-squares over their degrees of freedom is  $F$ . Dividing numerator and denominator by SST gives the last form.  $\square$

**Corollary 2.36.**  $R^2 = \frac{kF}{kF + n - k - 1}$ .

**Example 2.1.** A regression of a mutual fund's return on four predictors plus an intercept, over 105 monthly observations, reports  $F = 20.0$ ; the software failed before computing  $R^2$ . Since  $R^2$  depends on SSE and SST only through their ratio, and  $F$  determines that ratio, no further information is needed. With  $k = 4$  and  $n - k - 1 = 100$ ,

$$\frac{(\text{SST} - \text{SSE})/4}{\text{SSE}/100} = 20 \implies \frac{\text{SST} - \text{SSE}}{\text{SSE}} = 0.80 \implies \frac{\text{SST}}{\text{SSE}} = 1.80,$$

so  $R^2 = 1 - 1/1.80 = 0.444$ , as Corollary 2.36 also gives directly from  $kF = 80$ .

**Example 2.2.** Dental claims for  $n = 100$  participants. Actuary 1 fits  $Y = \beta + \varepsilon$ ; Actuary 2 fits  $Y = \beta_0 + \beta_1 \text{Sex} + \varepsilon$ , obtaining  $\text{SSE} = 250,000$  with  $\text{SST} = 490,000$ . Actuary 1's model is the null model, so its residual sum of squares is SST and no separate information about it is required. With  $k = 1$ ,

$$F = \frac{(\text{SST} - \text{SSE})/1}{\text{SSE}/(n - 2)} = \frac{240,000}{250,000/98} = 94.08 \quad \text{on } F_{1,98}.$$

Two checks: the prerequisites' corollary that the square of a  $t_\nu$  variable is  $F_{1,\nu}$  gives  $t = \sqrt{94.08} =$

9.70; and  $R^2 = 1 - 250,000/490,000 = 0.4898$  reproduces  $F$  through (14).

**Example 2.3.** (i) Observed  $y = (2, 5, 6, 8, 4)$ , fitted  $\hat{f} = (4, 3, 9, 3, 6)$ :  $\text{SSE} = 4 + 4 + 9 + 25 + 4 = 46$ . (ii) Simple linear regression with  $\sum_i e_i^2 = 230$  and  $R^2 = 0.64$ : from  $R^2 = 1 - \text{SSE}/\text{SST}$ ,  $\text{SST} = 230/0.36 = 638.89$ ;  $n$  is not needed.

## 7 The Case $k = 1$

FREES §2.1–2.5 · ISLR §3.1.

Simple linear regression is  $k = 1$ :  $\mathbf{X} = [\mathbf{1} \ \mathbf{x}]$ , and A4 reduces to the requirement that the  $x_i$  are not all equal.

**Definition 2.37** (Sample correlation). With  $s_x^2 = (n - 1)^{-1} \sum_i (x_i - \bar{x})^2$  and  $s_y^2$  likewise,

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}. \quad (15)$$

**Proposition 2.38** (Bounds).  $-1 \leq r \leq 1$ , with  $|r| = 1$  iff the points lie on a straight line.

**Proof.** With  $u_i = x_i - \bar{x}$  and  $v_i = y_i - \bar{y}$ , Cauchy–Schwarz gives  $(\sum_i u_i v_i)^2 \leq (\sum_i u_i^2)(\sum_i v_i^2)$ , i.e.  $r^2 \leq 1$ , with equality iff  $v_i = c u_i$  for all  $i$ .  $\square$

**Corollary 2.39** (Least squares estimates).

$$b_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = r \frac{s_y}{s_x}, \quad b_0 = \bar{y} - b_1 \bar{x}. \quad (16)$$

**Proof.** With  $\mathbf{X} = [\mathbf{1} \ \mathbf{x}]$  the normal equations (6) read

$$\begin{pmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{pmatrix} \begin{pmatrix} b_0 \\ b_1 \end{pmatrix} = \begin{pmatrix} \sum_i y_i \\ \sum_i x_i y_i \end{pmatrix}.$$

The first row is  $n b_0 + b_1 \sum_i x_i = \sum_i y_i$ , i.e.  $b_0 = \bar{y} - b_1 \bar{x}$ . Substituting into the second and using  $\sum_i x_i^2 - n \bar{x}^2 = \sum_i (x_i - \bar{x})^2$  and  $\sum_i x_i y_i - n \bar{x} \bar{y} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$  gives the stated  $b_1$ ; dividing numerator and denominator by  $n - 1$  turns it into  $r s_y / s_x$ .  $\square$

**Remark.**  $b_0 = \bar{y} - b_1\bar{x}$  says the fitted line passes through  $(\bar{x}, \bar{y})$ , which is the  $k = 1$  form of the residual identity  $\mathbf{1}'\mathbf{e} = 0$ .

**Corollary 2.40** (Variance of the slope).

$$\text{var}(b_1) = \frac{\sigma^2}{(n-1)s_x^2}, \quad \text{SE}(b_1) = \frac{s}{s_x\sqrt{n-1}}, \quad \text{var}(b_0) = \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2} \right). \quad (17)$$

**Proof.** For  $\mathbf{X} = [\mathbf{1} \ \mathbf{x}]$ ,

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_i x_i^2 \end{pmatrix}, \quad (\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{n \sum_i (x_i - \bar{x})^2} \begin{pmatrix} \sum_i x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix},$$

the determinant being  $n \sum_i x_i^2 - n^2\bar{x}^2 = n \sum_i (x_i - \bar{x})^2$ . Reading the diagonal of  $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ ,

$$\text{var}(b_1) = \frac{\sigma^2 n}{n \sum_i (x_i - \bar{x})^2} = \frac{\sigma^2}{(n-1)s_x^2}, \quad \text{var}(b_0) = \frac{\sigma^2 \sum_i x_i^2}{n \sum_i (x_i - \bar{x})^2} = \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2} \right),$$

the last using  $\sum_i x_i^2 = \sum_i (x_i - \bar{x})^2 + n\bar{x}^2$ . Replacing  $\sigma$  by  $s$  gives  $\text{SE}(b_1)$ . □

**Remark.**  $\text{SE}(b_1)$  falls as the predictor spreads out, so where the  $x_i$  are placed is a design choice: doubling  $s_x$  at fixed  $n$  and  $\sigma$  halves the standard error of the slope.

**Corollary 2.41** ( $R^2 = r^2$ ). With one explanatory variable and an intercept,  $R^2 = r^2$ .

**Proof.** By Corollary 2.39,  $\hat{y}_i - \bar{y} = b_1(x_i - \bar{x})$ , so  $\text{SSR} = b_1^2 \sum_i (x_i - \bar{x})^2 = b_1^2(n-1)s_x^2$ . Substituting  $b_1 = r s_y / s_x$  gives  $\text{SSR} = r^2(n-1)s_y^2 = r^2\text{SST}$ .  $\square$

**Remark.** So when  $k = 1$  the sign of the association is carried by  $r$  and its strength by  $R^2$ , and the two are equivalent. This fails for  $k > 1$ : there  $R^2$  is the squared correlation between  $y_i$  and  $\hat{y}_i$ , not between  $y$  and any one predictor. Note also that the overall  $F$  then tests the single slope, and  $F = t^2$  because the square of a  $t_\nu$  variable is  $F_{1,\nu}$ , which is the prerequisites' corollary on those two laws.

**Example 2.4** (The market model). FREES §2.7 fits  $r_{it} = \beta_0 + \beta_1 r_{mt} + \varepsilon_t$ , the return on a security against the return on a market index. Here  $\beta_1$  measures sensitivity to market movements, with  $\beta_1 > 1$  amplifying them, and by Corollary 2.41,  $R^2 = r^2$  is the proportion of the security's return

variation attributable to the market, its systematic share;  $1 - R^2$  is the diversifiable share.

## 8 Computation: Formulas Against Package Output

ISLR §3.6.3.

Data: `Boston` from `ISLR2`, with  $k = 3$ .

```
1 library(ISLR2)
2 fit <- lm(medv ~ lstat + age + rm, data = Boston)
3 X <- model.matrix(fit)                # n x (k+1), intercept first
4 y <- Boston$medv
5 e <- residuals(fit)
6 n <- nrow(X); k <- ncol(X) - 1L
7 c(n = n, k = k, df = n - k - 1L)
```

```
##    n    k  df
## 506    3 502
```

### 8.1 The estimator and its variance (§2, §4.1)

```
1 XtX <- t(X) %*% X
2 b_man <- drop(solve(XtX, t(X) %*% y))    # (X'X)^{-1} X'Y
```

```
3 cbind(manual = b_man, lm = coef(fit), diff = b_man - coef(fit))
```

```
##          manual          lm          diff
## (Intercept) -1.175311 -1.175311  1.722e-12
## lstat      -0.668513 -0.668513 -1.699e-14
## age        0.009091  0.009091  1.568e-15
## rm         5.019133  5.019133 -2.585e-13
```

**Remark.**  $(\mathbf{X}'\mathbf{X})^{-1}$  is never formed explicitly above: `solve(XtX, .)` solves the system. `lm` does not even form  $\mathbf{X}'\mathbf{X}$ , factoring  $\mathbf{X} = \mathbf{QR}$  instead, which avoids squaring the condition number. Equation (4) is how OLS is reasoned about, not how it is computed.

```
1 s2      <- sum(e^2) / (n - k - 1)           # MSE = SSE/(n-k-1)
2 V_man <- s2 * solve(XtX)                   # s^2 (X'X)^{-1}
3 c(max_abs_diff = max(abs(V_man - vcov(fit))),
4   s2_manual = s2, s2_lm = summary(fit)$sigma^2)
```

```
## max_abs_diff    s2_manual      s2_lm
##      5.240e-13      3.072e+01      3.072e+01
```

## 8.2 Projection, residuals, leverage (§3)

```

1 H <- X %*% solve(XtX) %*% t(X)           #  $H = X(X'X)^{-1}X'$ 
2 M <- diag(n) - H
3 c(symmetric = max(abs(H - t(H))),        #  $H' = H$ 
4   idempotent = max(abs(H %*% H - H)),     #  $H^2 = H$ 
5   HM_zero    = max(abs(H %*% M)),        #  $HM = 0$ 
6   MX_zero    = max(abs(M %*% X)))        #  $MX = 0$ 

```

```

## symmetric idempotent    HM_zero    MX_zero
## 2.082e-16  3.803e-15  3.803e-15  4.476e-12

```

```

1 c(tr_H = sum(diag(H)), k_plus_1 = k + 1,   #  $tr(H) = k+1$ 
2   tr_M = sum(diag(M)), n_minus_k_1 = n - k - 1)

```

```

##          tr_H      k_plus_1          tr_M n_minus_k_1
##          4         4          502         502

```

```

1 c(fitted    = max(abs(drop(H %*% y) - fitted(fit))), #  $\hat{Y} = HY$ 
2   resid     = max(abs(drop(M %*% y) - e)),           #  $e = MY$ 
3   Me_eq_e   = max(abs(drop(M %*% e) - e)),           #  $e$  already in  $C(X)$ -perp
4   leverage  = max(abs(diag(H) - hatvalues(fit))))    #  $h_{ii} = (i,i)$  entry of  $H$ 

```

```
##      fitted      resid  Me_eq_e  leverage
## 1.428e-12 1.450e-12 2.487e-14 3.775e-15
```

The identity  $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$  of (8) cannot be checked on real data,  $\boldsymbol{\varepsilon}$  being unobservable: it needs a design where  $\boldsymbol{\varepsilon}$  is set rather than observed.

```
1 c(Xt_e      = max(abs(t(X) %*% e)),      # X'e = 0
2   sum_e      = sum(e),                  # 1'e = 0 (needs the intercept)
3   yhat_t_e    = sum(fitted(fit) * e),    # Yhat'e = 0
4   h_bounds    = c(min(diag(H)), max(diag(H)))) # 0 <= h_ii <= 1
```

```
##      Xt_e      sum_e  yhat_t_e h_bounds1 h_bounds2
## 9.095e-13 4.233e-14 2.248e-12 2.052e-03 7.707e-02
```

Theorem 2.21 says  $\text{var}(e_i) = \sigma^2(1 - h_{ii})$ , so the standardised residual is  $e_i/(s\sqrt{1 - h_{ii}})$ ; this is what `rstandard` returns.

```
1 r_man <- e / (sqrt(s2) * sqrt(1 - hatvalues(fit)))
2 c(max_abs_diff = max(abs(r_man - rstandard(fit))),
3   sd_raw_e = sd(e), sd_std_e = sd(r_man))
```

```
## max_abs_diff      sd_raw_e      sd_std_e
##          0.000          5.526          1.005
```

## 8.3 Standard errors, $t$ and $p$ (§5)

```

1 se_man <- sqrt(diag(V_man))           # sqrt of diag( $s^2 (X'X)^{-1}$ )
2 t_man  <- coef(fit) / se_man          #  $\beta_j / se(\beta_j)$ 
3 p_man  <- 2 * pt(abs(t_man), df = n - k - 1, lower.tail = FALSE)
4 tab    <- coef(summary(fit))
5 round(cbind(se_man, se_lm = tab[, 2],
6             t_man,  t_lm  = tab[, 3],
7             p_man,  p_lm  = tab[, 4]), 8)

```

```

##              se_man  se_lm   t_man   t_lm p_man  p_lm
## (Intercept) 3.18192 3.18192 -0.3694 -0.3694 0.712 0.712
## lstat       0.05436 0.05436 -12.2985 -12.2985 0.000 0.000
## age        0.01122 0.01122  0.8106  0.8106 0.418 0.418
## rm         0.45431 0.45431 11.0479 11.0479 0.000 0.000

```

## 8.4 Goodness of fit (§6)

```

1 SSE <- sum(e^2)
2 SST <- sum((y - mean(y))^2)
3 SSR <- SST - SSE
4 c(SST = SST, SSR_plus_SSE = SSR + SSE,      #  $SST = SSR + SSE$ 

```

```
5 diff = SST - (SSR + SSE))
```

```
##          SST SSR_plus_SSE          diff
##      42716      42716              0
```

```
1 # anova() reports SEQUENTIAL sums of squares; the first k rows sum to SSR.
2 seq_ss <- anova(fit)[["Sum Sq"]]
3 c(SSR = SSR, sum_sequential = sum(seq_ss[1:k]),
4   SSE = SSE, anova_residual = seq_ss[k + 1L])
```

```
##          SSR sum_sequential          SSE anova_residual
##      27297      27297      15419      15419
```

```
1 R2      <- 1 - SSE / SST                                # coefficient of determination
2 adjR2 <- 1 - (n - 1) / (n - k - 1) * (1 - R2)
3 Fst     <- (SSR / k) / (SSE / (n - k - 1))              # MSR / MSE
4 c(R2 = R2,      R2_lm = summary(fit)$r.squared,
5   adj = adjR2,  adj_lm = summary(fit)$adj.r.squared,
6   F     = Fst,   F_lm  = summary(fit)$fstatistic[["value"]])
```

```
##          R2      R2_lm      adj      adj_lm          F          F_lm
##      0.6390      0.6390      0.6369      0.6369 296.2377 296.2377
```

```

1 c(R2_def      = R2,                                # 1 - SSE/SST
2   R2_cor2     = cor(y, fitted(fit))^2,              # squared correlation of y, yhat
3   R2_from_F   = k * Fst / (k * Fst + n - k - 1),
4   F_from_R2   = (R2 / k) / ((1 - R2) / (n - k - 1)))

```

```

##      R2_def      R2_cor2 R2_from_F F_from_R2
##      0.639      0.639      0.639    296.238

```

The overall  $F$  test compares the fitted model with the intercept-only model, which is what `anova` does when given both:

```

1 null_fit <- lm(medv ~ 1, data = Boston)
2 cmp <- anova(null_fit, fit)
3 c(F_manual = Fst, F_anova = cmp[["F"]][2],
4   SSE_null = sum(residuals(null_fit)^2), SST = SST, # null RSS equals SST
5   p_manual = pf(Fst, k, n - k - 1, lower.tail = FALSE))

```

```

##      F_manual      F_anova      SSE_null      SST      p_manual
##      2.962e+02      2.962e+02      4.272e+04      4.272e+04      1.203e-110

```

## 8.5 Likelihood, $\sigma^2$ and prediction (§5.2)

```

1 s2_mle <- SSE / n                                # the MLE divides by n, not n-k-1
2 c(s2_unbiased = s2, s2_mle = s2_mle,
3   ratio = s2_mle / s2, factor = (n - k - 1) / n)

```

```

## s2_unbiased      s2_mle      ratio      factor
##      30.7154      30.4726      0.9921      0.9921

```

Equation (10) evaluated at the MLE, against `logLik`; R's `logLik.lm` uses  $\hat{\sigma}_{\text{MLE}}^2$  and counts  $k + 2$  parameters (the  $k + 1$  coefficients and  $\sigma$ ), which is what makes the AIC agree.

```

1 ll_man <- -n/2 * log(2*pi) - n/2 * log(s2_mle) - SSE / (2 * s2_mle)
2 c(loglik_manual = ll_man, loglik_R = as.numeric(logLik(fit)),
3   npar_R       = attr(logLik(fit), "df"),
4   AIC_manual   = -2*ll_man + 2*(k + 2), AIC_R = AIC(fit),
5   BIC_manual   = -2*ll_man + log(n)*(k + 2), BIC_R = BIC(fit))

```

```

## loglik_manual      loglik_R      npar_R      AIC_manual
##      -1582          -1582          5          3175
##      AIC_R      BIC_manual      BIC_R
##      3175          3196          3196

```

```
1 x0 <- c(1, lstat = 10, age = 60, rm = 6.5)      # a new observation, intercept first
2 nd <- data.frame(lstat = 10, age = 60, rm = 6.5)
3 c(manual = sum(x0 * coef(fit)),
4   predict = unname(predict(fit, newdata = nd)))
```

```
## manual predict
##    25.31    25.31
```

## 8.6 The sampling theory by simulation (§4, §5.2)

Theorems 2.16, 2.17 and 2.27 are statements about repeated sampling. The design is held fixed, as the assumptions require, and only the errors are redrawn.

```
1 set.seed(1)
2 Xs <- X[1:60, ]                      # fixed design: n = 60, k = 3
3 ns <- nrow(Xs); ks <- ncol(Xs) - 1L
4 beta_true <- c(10, -0.5, 0.02, 3)
5 sigma <- 4; B <- 5000
6
7 bhat <- matrix(NA_real_, B, ks + 1L); sse <- numeric(B)
8 for (b in seq_len(B)) {
```

```

9   ysim <- drop(Xs %*% beta_true) + rnorm(ns, sd = sigma)
10  f <- lm.fit(Xs, ysim)
11  bhat[b, ] <- f$coefficients; sse[b] <- sum(f$residuals^2)
12 }

```

```

1  rbind(true = beta_true, mean_bhat = colMeans(bhat))           #  $E[bhat] = beta$ 

```

```

##           [,1]      [,2]      [,3]      [,4]
## true      10.000 -0.5000  0.02000  3.000
## mean_bhat   9.895 -0.4986  0.01977  3.016

```

```

1  round(cbind(empirical = apply(bhat, 2, var),                    #  $var(bhat)$ 
2           theory      = sigma^2 * diag(solve(t(Xs) %*% Xs))), 5)

```

```

##           empirical      theory
## (Intercept) 110.78282 110.66690
## lstat       0.02010   0.02031
## age         0.00069   0.00071
## rm          2.53789   2.54066

```

```

1  c(mean_SSE_over_s2 = mean(sse) / sigma^2, df = ns - ks - 1L, #  $chi$ -square moments

```

```
2 var_SSE_over_s2 = var(sse / sigma^2), two_df = 2*(ns - ks - 1L))
```

```
## mean_SSE_over_s2      df var_SSE_over_s2
##           56.1      56.0           108.7
##           two_df
##           112.0
```

With  $\epsilon$  known, (8) can also be checked directly:

```
1 set.seed(11)
2 eps <- rnorm(ns, sd = sigma)
3 Ms <- diag(ns) - Xs %*% solve(t(Xs) %*% Xs) %*% t(Xs)
4 e_sim <- lm.fit(Xs, drop(Xs %*% beta_true) + eps)$residuals
5 c(max_abs_diff = max(abs(e_sim - drop(Ms %*% eps))),           # e = M eps
6   sd_eps = sd(eps), sd_e = sd(e_sim))                          # e is not eps
```

```
## max_abs_diff      sd_eps      sd_e
##    1.128e-13    3.332e+00    3.291e+00
```

Independence of  $\hat{\beta}$  and SSE (Theorem 2.27, Step 3) shows up as zero correlation across replications:

```
1 round(cor(bhat, sse), 4)
```

```
##      [,1]\n## [1,] 0.0183\n## [2,] -0.0113\n## [3,] 0.0079\n## [4,] -0.0185
```

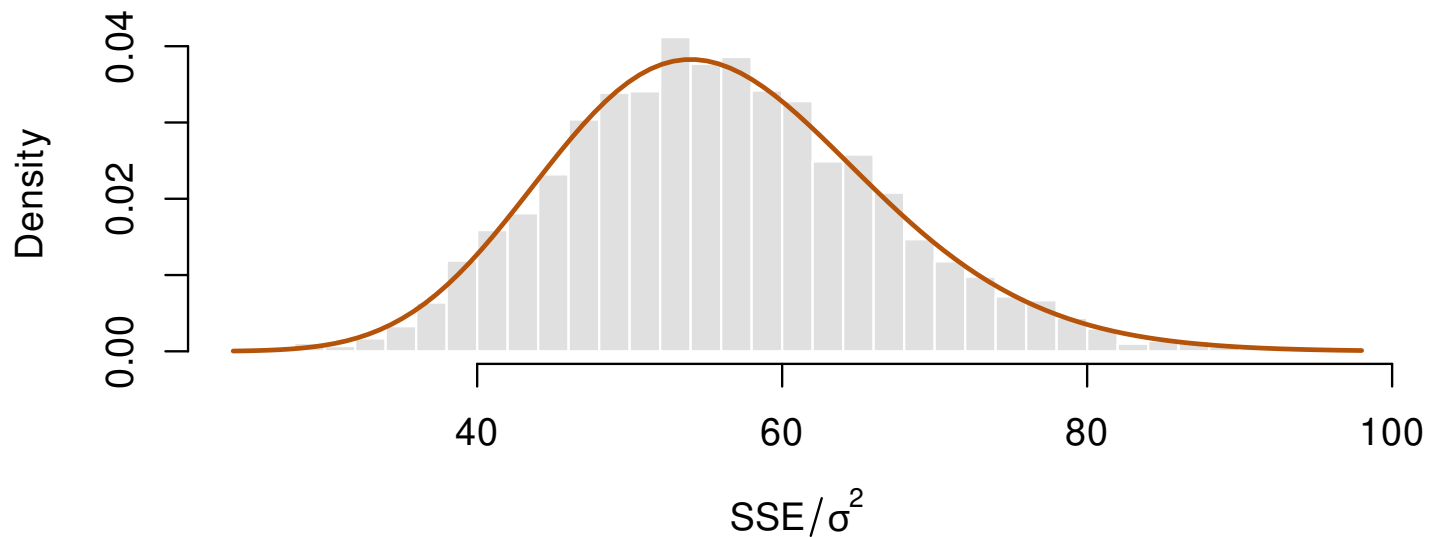


Figure 2.1: Simulated  $\text{SSE}/\sigma^2$  against the  $\chi_{56}^2$  density (Theorem 2.27),  $B = 5000$  replications on a fixed design.

## 8.7 The case $k = 1$ (§7)

```

1 xs <- Boston$lstat                                # a single predictor
2 slr <- lm(medv ~ lstat, data = Boston)
3 r <- cor(xs, y); sx <- sd(xs); sy <- sd(y)

1 b1 <- r * sy / sx                                # r s_y / s_x
2 b0 <- mean(y) - b1 * mean(xs)                    # ybar - b1 xbar
3 cbind(manual = c(b0, b1), lm = coef(slr), diff = c(b0, b1) - coef(slr))

##          manual      lm      diff
## (Intercept)  34.55 34.55  0.00e+00
## lstat        -0.95 -0.95 -2.22e-16

1 s_slr <- summary(slr)$sigma
2 se_b1 <- s_slr / (sx * sqrt(n - 1))              # s / (s_x sqrt(n-1))
3 se_b0 <- s_slr * sqrt(1/n + mean(xs)^2 / ((n - 1) * sx^2))

```

```
4 cbind(manual = c(se_b0, se_b1), lm = coef(summary(slr))[, "Std. Error"])
```

```
##              manual      lm
## (Intercept) 0.56263 0.56263
## lstat       0.03873 0.03873
```

```
1 c(R2_lm      = summary(slr)$r.squared,
2   r_squared  = r^2,                      #  $R^2 = r^2$  when  $k = 1$ 
3   cor_y_ymat = cor(y, fitted(slr))^2,
4   t_squared  = coef(summary(slr))["lstat", "t value"]^2,
5   F_lm       = summary(slr)$fstatistic[["value"]])
```

```
##      R2_lm  r_squared cor_y_ymat  t_squared      F_lm
##      0.5441    0.5441    0.5441   601.6179   601.6179
```

The market model of Example 2.4, simulated from a known  $\beta_1$  so that the regression can be checked against the truth:

```
1 set.seed(1)
2 Tn <- 240                      # 20 years of monthly returns
3 rmk <- rnorm(Tn, mean = 0.007, sd = 0.045)  # market return
4 beta_true <- 1.30; sd_idio <- 0.030
```

```

5 ri <- 0.001 + beta_true * rmk + rnorm(Tn, sd = sd_idio)
6 capm <- lm(ri ~ rmk)
7 c(beta_true = beta_true, beta_hat = coef(capm)[["rmk"]],
8   in_CI = as.numeric(beta_true >= confint(capm)[["rmk", 1] &
9                     beta_true <= confint(capm)[["rmk", 2]]),
10   R2 = summary(capm)$r.squared,
11   systematic_share = beta_true^2 * var(rmk) / (beta_true^2 * var(rmk) + sd_idio^2))

```

```

##          beta_true          beta_hat          in_CI
##          1.3000          1.3736          1.0000
##          R2 systematic_share
##          0.7946          0.7790

```

## 9 Pitfalls

### 9.1 Reading the residual identities as diagnostics

Theorem 2.11 holds for every fit. A straight line through quadratic data satisfies all three identities exactly.

```

1 set.seed(2)
2 x <- seq(-3, 3, length.out = 80)

```

```

3 yq <- x^2 + rnorm(80, sd = 0.3)           # truly quadratic
4 bad <- lm(yq ~ x)                         # a straight line is fitted anyway
5 c(sum_resid = sum(residuals(bad)),        # 1'e = 0 regardless of fit
6   cor_x_resid = cor(x, residuals(bad)),   # X'e = 0 regardless of fit
7   R2          = summary(bad)$r.squared)

```

```

##      sum_resid cor_x_resid          R2
## -2.126e-14    4.573e-17    1.405e-04

```

**Diagnostic.** Plot residuals against fitted values and look for curvature, which the identities do not forbid. **Fix.** Add the missing term; here, a quadratic one.

```

1 good <- lm(yq ~ x + I(x^2))
2 c(R2_linear = summary(bad)$r.squared, R2_quadratic = summary(good)$r.squared,
3   SSE_linear = sum(residuals(bad)^2), SSE_quadratic = sum(residuals(good)^2))

```

```

##      R2_linear  R2_quadratic    SSE_linear SSE_quadratic
##      0.0001405      0.9855337    620.2354128      8.9737812

```

## 9.2 Treating $R^2$ as a selection criterion

By Theorem 2.33,  $R^2$  rises when pure noise is added.

```

1 set.seed(3)
2 d <- Boston[1:60, ]
3 m1 <- lm(medv ~ lstat, data = d)
4 noise <- matrix(rnorm(60 * 20), 60, 20)           # 20 columns of noise
5 m2 <- lm(d$medv ~ d$lstat + noise)
6 c(R2_1 = summary(m1)$r.squared,      R2_2 = summary(m2)$r.squared,
7   adj1 = summary(m1)$adj.r.squared, adj2 = summary(m2)$adj.r.squared)

```

```

##      R2_1      R2_2      adj1      adj2
## 0.5991 0.6988 0.5922 0.5324

```

**Diagnostic.**  $R^2$  rises while  $\bar{R}^2$  falls. **Fix.** Compare on a penalised criterion ( $C_p$ , AIC, BIC) or on out-of-sample error.

## 9.3 $R^2$ without an intercept

Theorem 2.30 requires  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . Without it R switches to an uncentered definition of  $R^2$ , which is not comparable with the usual one.

```

1 m_int <- lm(medv ~ lstat, data = Boston)
2 m_no  <- lm(medv ~ lstat - 1, data = Boston)

```

```

3 c(R2_with = summary(m_int)$r.squared, R2_without = summary(m_no)$r.squared,
4   SSE_with = sum(residuals(m_int)^2), SSE_without = sum(residuals(m_no)^2),
5   sum_resid_without = sum(residuals(m_no)))

```

```

##           R2_with      R2_without      SSE_with
##           0.5441      0.4486      19472.3814
##   SSE_without sum_resid_without
##   165199.2060      4217.3843

```

**Diagnostic.** The no-intercept fit has the larger SSE and the larger reported  $R^2$ ; its residuals do not sum to zero, so Theorem 2.30 does not apply. **Fix.** Retain the intercept; compare models on SSE or an information criterion.

## 10 Exercises

**Exercise 2.1.** (a) Show  $\text{rank}(\mathbf{X}'\mathbf{X}) = \text{rank}(\mathbf{X})$  by proving the two matrices have the same null space, and say where Theorem 2.5 uses it. (b) Show  $\mathbf{X}'\mathbf{X}$  is always positive semi-definite, and positive definite exactly under A4. (c) Prove  $\mathbf{M} = \mathbf{I} - \mathbf{H}$  is symmetric idempotent and deduce  $\text{tr}(\mathbf{M}) = n - k - 1$  without quoting Theorem 2.10(iv).

**Solution.** (a) If  $\mathbf{X}\mathbf{v} = \mathbf{0}$  then  $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$ . Conversely  $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$  gives  $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 = 0$ , hence  $\mathbf{X}\mathbf{v} = \mathbf{0}$ . The null spaces coincide, and by rank–nullity applied to two matrices with  $k + 1$  columns, so do the ranks. Step 3 uses it: A4 gives  $\text{rank}(\mathbf{X}) = k + 1$ , and only through this equality does it make  $\mathbf{X}'\mathbf{X}$  invertible.

(b)  $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 \geq 0$  for every  $\mathbf{v}$ , with no assumption. It is strict for every  $\mathbf{v} \neq \mathbf{0}$  iff  $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$ , which by rank–nullity is A4. This is Step 4 of the same proof: the condition that makes  $\hat{\boldsymbol{\beta}}$  exist makes the Hessian positive definite, so nothing extra is needed for a minimum rather than a stationary point.

(c)  $\mathbf{M}' = \mathbf{I} - \mathbf{H}' = \mathbf{M}$  and  $\mathbf{M}^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{M}$ , using  $\mathbf{H}' = \mathbf{H}$  and  $\mathbf{H}^2 = \mathbf{H}$ . By linearity  $\text{tr}(\mathbf{M}) = n - \text{tr}(\mathbf{H})$ , and  $\mathbf{H}$  idempotent gives  $\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H})$  by Lemma 2.9. Finally  $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ : every column of  $\mathbf{H}$  lies in  $\mathcal{C}(\mathbf{X})$ , while  $\mathbf{H}\mathbf{X} = \mathbf{X}$  puts every column of  $\mathbf{X}$  in  $\mathcal{C}(\mathbf{H})$ . Hence  $\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{X}) = k + 1$  by A4. ■

**Exercise 2.2.** The remark after Theorem 2.10 asserts that symmetry, idempotency and  $\mathcal{C}(\mathbf{P}) = \mathcal{C}(\mathbf{X})$  characterise  $\mathbf{H}$ . Prove it. Which of the three fails for  $\mathbf{P} = \mathbf{H} + \mathbf{A}$  with  $\mathbf{A}$  antisymmetric and  $\mathbf{A}\mathbf{X} = \mathbf{0}$ ?

**Solution.** A symmetric idempotent  $\mathbf{P}$  acts as the identity on  $\mathcal{C}(\mathbf{P})$  and kills  $\mathcal{C}(\mathbf{P})^\perp$ : if  $\mathbf{v} = \mathbf{P}\mathbf{w}$  then  $\mathbf{P}\mathbf{v} = \mathbf{P}^2\mathbf{w} = \mathbf{v}$ , and by symmetry  $\mathcal{N}(\mathbf{P}) = \mathcal{N}(\mathbf{P}') = \mathcal{C}(\mathbf{P})^\perp$ . Both  $\mathbf{P}$  and  $\mathbf{H}$  are symmetric idempotent with column space  $\mathcal{C}(\mathbf{X})$ , so both are the identity on  $\mathcal{C}(\mathbf{X})$  and zero on  $\mathcal{C}(\mathbf{X})^\perp$ . Since  $\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^\perp$  they agree on a spanning set, so  $\mathbf{P} = \mathbf{H}$ .

For the perturbation,  $\mathbf{A}\mathbf{X} = \mathbf{0}$  gives  $\mathbf{A}\mathbf{H} = \mathbf{A}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{0}$  and  $\mathbf{H}\mathbf{A} = (\mathbf{A}'\mathbf{H})' = (-\mathbf{A}\mathbf{H})' = \mathbf{0}$ , so  $\mathbf{P}^2 = \mathbf{H} + \mathbf{A}^2$  and idempotency requires  $\mathbf{A}^2 = \mathbf{0}$ ; for antisymmetric  $\mathbf{A}$  that forces  $\text{tr}(\mathbf{A}\mathbf{A}') = -\text{tr}(\mathbf{A}^2) = 0$ , hence  $\mathbf{A} = \mathbf{0}$ . Symmetry fails first:  $\mathbf{P}' = \mathbf{H} - \mathbf{A} \neq \mathbf{P}$ . Dropping symmetry alone leaves an **oblique** projection onto  $\mathcal{C}(\mathbf{X})$ , which reproduces  $\mathcal{C}(\mathbf{X})$  but does not minimise  $\|\mathbf{Y} - \tilde{\mathbf{Y}}\|$ .



**Exercise 2.3** (ISLR §3.7, exercise 5). For regression through the origin,  $\hat{\beta} = \sum_i x_i y_i / \sum_j x_j^2$  and  $\hat{y}_i = x_i \hat{\beta}$ . Write  $\hat{y}_i = \sum_j a_j y_j$  and identify  $a_j$ . Show the same statement for the general model is  $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ , and that the weights sum to 1 when an intercept is present.

**Solution.** Substituting,

$$\hat{y}_i = x_i \frac{\sum_j x_j y_j}{\sum_l x_l^2} = \sum_j a_j y_j, \quad a_j = \frac{x_i x_j}{\sum_l x_l^2}.$$

So the fitted values are linear combinations of the responses with weights determined by  $\mathbf{X}$  alone. For the general model  $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ , so  $\hat{Y}_i = \sum_j h_{ij} Y_j$  and the weight of  $Y_j$  in  $\hat{Y}_i$  is  $h_{ij}$ . The single-regressor case is  $\mathbf{H} = \mathbf{xx}'/(\mathbf{x}'\mathbf{x})$ , whose  $(i, j)$  entry is the  $a_j$  above. With an intercept,  $\mathbf{H}\mathbf{1} = \mathbf{1}$  because  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , so  $\sum_j h_{ij} = 1$  for every  $i$ : each fitted value is a weighted average of the responses, with weights summing to one but not necessarily non-negative. The weight an observation gives itself is  $h_{ii}$ , its leverage (Definition 2.12), which is why  $\partial \hat{Y}_i / \partial Y_i = h_{ii}$ . ■

**Exercise 2.4.** Derive  $\text{var}(e_i) = \sigma^2(1 - h_{ii})$  and explain why a high-leverage observation has a less variable residual. Deduce  $\text{cov}(e_i, e_j) = -\sigma^2 h_{ij}$ , and compute the residual correlation implied by  $h_{ij} = -0.02$ ,  $h_{ii} = h_{jj} = 0.05$ .

**Solution.** From  $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$  and Lemma 2.15,  $\text{var}(\mathbf{e}) = \mathbf{M}(\sigma^2\mathbf{I})\mathbf{M}' = \sigma^2\mathbf{M}$  by symmetry and idempotency; the  $(i, i)$  entry of  $\mathbf{I} - \mathbf{H}$  is  $1 - h_{ii}$  and the  $(i, j)$  entry is  $-h_{ij}$ .

A high-leverage point has extreme predictor values, so the fitted surface is pulled toward it and its residual has little freedom; as  $h_{ii} \rightarrow 1$  the fit passes through the point and the residual is identically zero.

For the correlation,

$$\text{cor}(e_i, e_j) = \frac{-\sigma^2 h_{ij}}{\sigma^2 \sqrt{(1 - h_{ii})(1 - h_{jj})}} = \frac{0.02}{0.95} = 0.0211.$$

Since  $|h_{ij}| \leq \sqrt{h_{ii}h_{jj}}$  and the average leverage is  $(k + 1)/n$ , the residual correlations are of order  $(k + 1)/n$ : negligible when  $n$  is large relative to  $k$ , and not otherwise. Any test treating residuals as independent carries that error before model adequacy is considered. ■

**Exercise 2.5.** With an intercept, prove  $h_{ii} \geq 1/n$ , with equality iff  $\mathbf{x}_i$  sits at the centroid of the

design. For  $k = 1$  derive

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}$$

and check it against `hatvalues` for `medv` on `lstat`.

**Solution.** Let  $\mathbf{H}_0 = \frac{1}{n}\mathbf{1}\mathbf{1}'$ , with all diagonal entries  $1/n$ . By Lemma 2.34,  $\mathbf{H} - \mathbf{H}_0$  is symmetric idempotent, hence positive semi-definite, so  $h_{ii} - 1/n = \mathbf{e}_i'(\mathbf{H} - \mathbf{H}_0)\mathbf{e}_i = \|(\mathbf{H} - \mathbf{H}_0)\mathbf{e}_i\|^2 \geq 0$  for the  $i$ -th coordinate vector. Equality forces  $(\mathbf{H} - \mathbf{H}_0)\mathbf{e}_i = \mathbf{0}$ , that is, the  $i$ -th row of  $\mathbf{H}$  is constant at  $1/n$ , which happens exactly when  $\mathbf{x}_i$  is at the centroid.

For  $k = 1$ , with  $S_{xx} = \sum_j (x_j - \bar{x})^2$  and  $\mathbf{x}_i' = (1, x_i)$ , the explicit inverse of Corollary 2.40 gives

$$h_{ii} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i = \frac{\sum_j x_j^2 - 2n\bar{x}x_i + nx_i^2}{nS_{xx}} = \frac{S_{xx} + n(x_i - \bar{x})^2}{nS_{xx}} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}},$$

using  $\sum_j x_j^2 = S_{xx} + n\bar{x}^2$ . Leverage in simple regression is an exact quadratic in the distance from  $\bar{x}$ , with minimum  $1/n$ .

```
1 Sxx <- sum((xs - mean(xs))^2)
```

```
# xs is Boston$lstat
```

```

2 h1 <- 1/n + (xs - mean(xs))^2 / Sxx
3 c(max_abs_diff = max(abs(h1 - hatvalues(slr))), one_over_n = 1/n,
4   min_h = min(h1), max_h = max(h1),
5   lstat_at_min = xs[which.min(h1)], mean_lstat = mean(xs))

```

```

## max_abs_diff    one_over_n        min_h        max_h
##      2.515e-17      1.976e-03      1.976e-03      2.687e-02
## lstat_at_min    mean_lstat
##      1.264e+01      1.265e+01

```

**Exercise 2.6.** Show  $h_{ii} = 1$  forces  $h_{ij} = 0$  for every  $j \neq i$  and  $e_i = 0$ . Construct a design where this happens, fit it, and say what the model then estimates.

**Solution.** From the proof of Theorem 2.13(i),  $h_{ii} = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$ ; at  $h_{ii} = 1$  the first two terms are equal, so every  $h_{ij}$  vanishes. Then  $\hat{Y}_i = \sum_j h_{ij} Y_j = Y_i$  and  $e_i = 0$ , consistently with  $\text{var}(e_i) = \sigma^2(1 - h_{ii}) = 0$ .

An indicator that is 1 for observation  $i$  alone does it: the column lets the fit reproduce  $Y_i$  exactly at no cost elsewhere.

```

1 dd <- data.frame(y = c(3, 5, 4, 9, 6), xv = 1:5,
2                   dum = as.numeric(1:5 == 4))
3 md <- lm(y ~ xv + dum, data = dd); mn <- lm(y ~ xv, data = dd)
4 rbind(hat = round(hatvalues(md), 4), resid = round(residuals(md), 10),
5        resid_no_dummy = round(residuals(mn), 4))

```

```

##           1         2         3  4         5
## hat      0.6 0.3143  0.2571 1.0  0.82860
## resid    -0.4 0.9714 -0.6571 0.0  0.08571
## resid_no_dummy -0.4 0.6000 -1.4000 2.6 -1.40000

```

```

1 c(sum_h = sum(hatvalues(md)), k_plus_1 = 3,
2   SSE_with = sum(residuals(md)^2), SSE_without = sum(residuals(mn)^2))

```

```

##      sum_h    k_plus_1    SSE_with SSE_without
##      3.000        3.000        1.543        11.200

```

The leverages sum to  $k + 1$  as Theorem 2.13(ii) requires, and the fourth residual is exactly zero. The model then fits a line to the other four observations plus a parameter absorbing  $Y_i$  whole: observation  $i$  contributes nothing to the slope and nothing to SSE, and the residual degrees of freedom fall by one to record it. A singleton indicator is a formal deletion of an observation, which

is worth recognising when one appears by accident as a rare factor level. ■

**Exercise 2.7.** Prove  $SST = SSR + SSE$ , stating where the intercept is used, and give a design for which the identity fails.

**Solution.** The proof is Theorem 2.30. The intercept is used twice: Step 2 needs  $\bar{e} = 0$  so that  $\mathbf{M}_0 \mathbf{e} = \mathbf{e}$ , and Step 4 needs  $\mathbf{e}' \mathbf{1} = 0$ . Both are Theorem 2.11(ii), which holds only because  $\mathbf{1}$  is a column of  $\mathbf{X}$ .

Take  $n = 2$ ,  $\mathbf{X} = (1, 2)'$  without an intercept,  $\mathbf{Y} = (3, 3)'$ . Then  $\hat{\beta} = \mathbf{X}'\mathbf{Y}/\mathbf{X}'\mathbf{X} = 9/5 = 1.8$ ,  $\hat{\mathbf{Y}} = (1.8, 3.6)'$  and  $\mathbf{e} = (1.2, -0.6)'$ , so  $\sum_i e_i = 0.6 \neq 0$ . Here  $SST = 0$  while  $SSR = 1.62$  and  $SSE = 1.8$ , so  $SSR + SSE = 3.42 \neq 0 = SST$ . ■

**Exercise 2.8.** Prove that in any model with an intercept,  $R^2$  equals the squared sample correlation between  $Y_i$  and  $\hat{Y}_i$ . Deduce Corollary 2.41 as the case  $k = 1$ , and explain why  $R^2$  is not the squared correlation with any single predictor when  $k > 1$ .

**Solution.** Since  $\widehat{\bar{Y}} = \bar{Y}$  (Theorem 2.30, Step 1), the covariance numerator is

$$\sum_i (Y_i - \bar{Y})(\hat{Y}_i - \bar{Y}) = \hat{\mathbf{Y}}' \mathbf{M}_0 \mathbf{Y} = \hat{\mathbf{Y}}' \mathbf{M}_0 (\hat{\mathbf{Y}} + \mathbf{e}) = \text{SSR} + \hat{\mathbf{Y}}' \mathbf{e} = \text{SSR},$$

using  $\mathbf{M}_0 \mathbf{e} = \mathbf{e}$  and Theorem 2.11(iii). The two sums of squares in the denominator are SST and SSR, so  $\text{cor}(Y, \hat{Y})^2 = \text{SSR}^2 / (\text{SST} \cdot \text{SSR}) = R^2$ . This is the check `R2_cor2` of §8.4.

For  $k = 1$ ,  $\hat{Y}_i = b_0 + b_1 x_i$  is affine in  $x_i$  and correlation is invariant under an affine map up to sign, so  $\text{cor}(Y, \hat{Y})^2 = r^2$ : Corollary 2.41.

For  $k > 1$ ,  $\hat{\mathbf{Y}}$  combines all  $k$  predictors.  $R^2$  can be large while every individual  $|\text{cor}(Y, X_j)|$  is small, if the predictors combine, and barely above  $\max_j \text{cor}(Y, X_j)^2$  when they are near-duplicates.



**Exercise 2.9.** Let  $\mathbf{X}_+ = [\mathbf{X} \ \mathbf{z}]$  with  $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$  and  $\mathbf{M} = \mathbf{I} - \mathbf{H}$ . (a) Show the projection onto  $\mathcal{C}(\mathbf{X}_+)$  minus that onto  $\mathcal{C}(\mathbf{X})$  is  $\mathbf{Mz}(\mathbf{z}'\mathbf{Mz})^{-1}\mathbf{z}'\mathbf{M}$ , of rank one. (b) Deduce  $\text{SSE} - \text{SSE}_+ = (\mathbf{z}'\mathbf{MY})^2 / (\mathbf{z}'\mathbf{Mz})$ . (c) Show the coefficient on  $\mathbf{z}$  is  $\hat{\gamma} = \mathbf{z}'\mathbf{MY} / (\mathbf{z}'\mathbf{Mz})$  with  $\text{var}(\hat{\gamma}) = \sigma^2 / (\mathbf{z}'\mathbf{Mz})$ . (d) Conclude  $t^2 = (\text{SSE} - \text{SSE}_+) / s_+^2$ . (e) Verify all four by adding `rm` to `medv` on `lstat` and `age`.

**Solution.** (a)  $\mathcal{C}(\mathbf{X}_+) = \mathcal{C}(\mathbf{X}) \oplus \text{span}(\mathbf{Mz})$ : the two pieces are orthogonal because  $\mathbf{X}'\mathbf{M} = \mathbf{0}$ , both lie in  $\mathcal{C}(\mathbf{X}_+)$  because  $\mathbf{Mz} = \mathbf{z} - \mathbf{Hz}$ , and their dimensions add to  $k + 2$ . The projection onto an orthogonal direct sum is the sum of the projections, and the projection onto  $\text{span}(\mathbf{Mz})$  is  $\mathbf{P}_z = \mathbf{Mz}(\mathbf{z}'\mathbf{Mz})^{-1}\mathbf{z}'\mathbf{M}$ , using  $\|\mathbf{Mz}\|^2 = \mathbf{z}'\mathbf{Mz}$ . So  $\mathbf{H}_+ = \mathbf{H} + \mathbf{P}_z$  with  $\text{rank}(\mathbf{P}_z) = 1$ .

(b)  $\text{SSE} - \text{SSE}_+ = \mathbf{Y}'(\mathbf{I} - \mathbf{H})\mathbf{Y} - \mathbf{Y}'(\mathbf{I} - \mathbf{H}_+)\mathbf{Y} = \mathbf{Y}'\mathbf{P}_z\mathbf{Y} = (\mathbf{z}'\mathbf{MY})^2/(\mathbf{z}'\mathbf{Mz})$ .

(c) Write  $\hat{\mathbf{Y}}_+ = \mathbf{X}\hat{\boldsymbol{\beta}}_+ + \mathbf{z}\hat{\gamma}$  and apply  $\mathbf{M}$ :  $\mathbf{M}\hat{\mathbf{Y}}_+ = \hat{\gamma}\mathbf{Mz}$ . But  $\mathbf{M}\hat{\mathbf{Y}}_+ = \mathbf{M}(\mathbf{H} + \mathbf{P}_z)\mathbf{Y} = \mathbf{P}_z\mathbf{Y} = \mathbf{Mz}(\mathbf{z}'\mathbf{MY})/(\mathbf{z}'\mathbf{Mz})$ , since  $\mathbf{MH} = \mathbf{0}$  and  $\mathbf{MP}_z = \mathbf{P}_z$ . Comparing gives  $\hat{\gamma}$ ; it is a fixed linear map of  $\mathbf{Y}$ , so Lemma 2.15 gives  $\text{var}(\hat{\gamma}) = \sigma^2\mathbf{z}'\mathbf{M}\mathbf{Mz}/(\mathbf{z}'\mathbf{Mz})^2 = \sigma^2/(\mathbf{z}'\mathbf{Mz})$ .

(d)  $t^2 = \hat{\gamma}^2(\mathbf{z}'\mathbf{Mz})/s_+^2 = (\mathbf{z}'\mathbf{MY})^2/\{(\mathbf{z}'\mathbf{Mz})s_+^2\}$ , which is (b) divided by  $s_+^2$ .

```

1 m_old <- lm(medv ~ lstat + age, data = Boston)
2 X0 <- model.matrix(m_old)
3 M0 <- diag(n) - X0 %*% solve(crossprod(X0)) %*% t(X0)
4 z <- Boston$rm
5 zMz <- drop(t(z) %*% M0 %*% z); zMy <- drop(t(z) %*% M0 %*% y)
6 s2p <- sum(residuals(fit)^2)/(n - 4)
7 c(gamma_formula = zMy/zMz,          gamma_lm = coef(fit)[["rm"]],
8   se_formula    = sqrt(s2p/zMz),    se_lm    = coef(summary(fit))["rm", 2],

```

```

9  SSE_drop      = sum(residuals(m_old)^2) - sum(residuals(fit)^2),
10 quad_form     = zMy^2/zMz,
11 t_squared     = coef(summary(fit))["rm", 3]^2,
12 drop_over_s2  = (sum(residuals(m_old)^2) - sum(residuals(fit)^2))/s2p)

```

```

## gamma_formula      gamma_lm      se_formula      se_lm
##          5.0191          5.0191          0.4543          0.4543
##      SSE_drop      quad_form      t_squared      drop_over_s2
##      3749.0035      3749.0035      122.0562      122.0562

```

Nothing here needs normality: (a) to (c) are algebra and Lemma 2.15, and only reading  $t$  as a  $t$  distribution needs A5. ■

**Exercise 2.10.** The remark after Theorem 2.33 says  $\bar{R}^2$  rises when a variable with  $|t| > 1$  is added. Prove it, and verify on **Boston** with noise columns whose  $t$  falls either side of 1.

**Solution.**  $\bar{R}^2 = 1 - s^2/\{\text{SST}/(n-1)\}$  and SST does not change, so  $\bar{R}^2$  rises iff  $s^2$  falls. With  $s_+^2 = \text{SSE}_+/(n-k-2)$  and  $s^2 = \text{SSE}/(n-k-1)$ ,

$$s_+^2 < s^2 \iff \text{SSE}_+(n-k-1) < \text{SSE}(n-k-2) \iff \text{SSE} - \text{SSE}_+ > \frac{\text{SSE}_+}{n-k-2} = s_+^2,$$

subtracting  $\text{SSE}_+(n - k - 2)$  from both sides. By Exercise 2.9(d) the left side is  $t^2 s_+^2$ , so the condition is exactly  $t^2 > 1$ .

```

1 base <- lm(medv ~ lstat + age, data = Boston)
2 t(sapply(c(202, 203, 204, 205), function(sd) {
3   set.seed(sd); zz <- rnorm(n)
4   mb <- lm(medv ~ lstat + age + zz, data = Boston)
5   c(seed = sd, t = coef(summary(mb))["zz", 3],
6     adjR2_before = summary(base)$adj.r.squared,
7     adjR2_after  = summary(mb)$adj.r.squared)
8 })))

```

| ##      | seed | t       | adjR2_before | adjR2_after |
|---------|------|---------|--------------|-------------|
| ## [1,] | 202  | 0.2142  | 0.5495       | 0.5486      |
| ## [2,] | 203  | 0.7455  | 0.5495       | 0.5491      |
| ## [3,] | 204  | 1.0384  | 0.5495       | 0.5496      |
| ## [4,] | 205  | -1.5930 | 0.5495       | 0.5509      |

The turn falls between  $|t| = 0.75$  and  $|t| = 1.04$ . A variable is retained by  $\bar{R}^2$  on evidence far weaker than  $|t| > 2$ , which is why  $\bar{R}^2$  selects models that are too large. ■

**Exercise 2.11** (FREES exercise 3.1). A dataset of  $n = 100$  observations has  $s_y = 80$ ; a regression on three explanatory variables gives  $s = 50$ . Reconstruct the whole ANOVA table, then compute  $R^2$ ,  $\bar{R}^2$  and  $F$ , and test model adequacy at 5%. State the effect on each quantity of adding a fourth predictor that is pure noise.

**Solution.** Two mean squares determine everything:  $SST = (n - 1)s_y^2$  and  $SSE = (n - k - 1)s^2$ , and  $SSR = SST - SSE$  by Theorem 2.30.

```

1 nA <- 100; kA <- 3; syA <- 80; sA <- 50
2 SSTa <- (nA - 1)*syA^2; SSEa <- (nA - kA - 1)*sA^2; SSRa <- SSTa - SSEa
3 c(SST = SSTa, SSR = SSRa, SSE = SSEa,
4   MSR = SSRa/kA, MSE = sA^2, F = (SSRa/kA)/sA^2,
5   F_crit_5pct = qf(0.95, kA, nA - kA - 1),
6   R2 = SSRa/SSTa, adjR2 = 1 - sA^2/syA^2)
```

| ## | SST         | SSR         | SSE         | MSR         | MSE       |
|----|-------------|-------------|-------------|-------------|-----------|
| ## | 633600.0000 | 393600.0000 | 240000.0000 | 131200.0000 | 2500.0000 |
| ## | F           | F_crit_5pct | R2          | adjR2       |           |
| ## | 52.4800     | 2.6994      | 0.6212      | 0.6094      |           |

The table is

| Source     | Sum of squares | df | Mean square |
|------------|----------------|----|-------------|
| Regression | 393,600        | 3  | 131,200     |
| Error      | 240,000        | 96 | 2,500       |
| Total      | 633,600        | 99 |             |

$F = 52.48$  against a 5% critical value of 2.70, so the three predictors are jointly significant. Note  $\bar{R}^2 = 1 - s^2/s_y^2$  exactly, which is the definition in (13) rearranged, and needs no sums of squares at all.

Adding a noise predictor: SST is unchanged, since it does not involve  $\mathbf{X}$ ; SSE falls weakly and SSR rises weakly, so  $R^2$  rises (Theorem 2.33);  $s$  and  $\bar{R}^2$  move according to whether the new  $|t|$  exceeds 1 (Exercise 2.10), which for a null variable on 95 degrees of freedom happens with probability 0.32; and  $F$  falls, since the numerator degrees of freedom rise while SSR hardly moves. ■

**Exercise 2.12** (FREES exercise 4.1). (a) Establish  $F = \frac{R^2}{1 - R^2} \cdot \frac{n - k - 1}{k}$  from Definition 2.14. (b) Compute  $F$  and test model adequacy at 5% for  $k = 5$ ,  $R^2 = 0.20$ , first at  $n = 40$  and then at

$n = 400$ . Explain the difference in one sentence.

**Solution.** (a)  $F = \text{MSR}/\text{MSE} = (\text{SSR}/k)/\{\text{SSE}/(n - k - 1)\}$ . Dividing numerator and denominator by SST and using  $\text{SSR}/\text{SST} = R^2$ ,  $\text{SSE}/\text{SST} = 1 - R^2$  (Theorem 2.30) gives the display, which is (14).

```
1 t(sapply(c(40, 400), function(nn) {
2   R2v <- 0.20; kv <- 5
3   Fv <- R2v/(1 - R2v) * (nn - kv - 1)/kv
4   c(n = nn, F = Fv, crit_5pct = qf(0.95, kv, nn - kv - 1),
5     p_value = pf(Fv, kv, nn - kv - 1, lower.tail = FALSE))
6 })))
```

```
##          n      F crit_5pct  p_value
## [1,]   40   1.7      2.494 1.613e-01
## [2,]  400  19.7      2.237 1.569e-17
```

(b) At  $n = 40$ ,  $F = 1.70$  against a critical value of 2.49: not significant. At  $n = 400$ ,  $F = 19.7$  against 2.24: overwhelmingly significant. The same  $R^2$  is decisive or worthless depending on  $n$ , because  $F$  is  $R^2/(1 - R^2)$  scaled by the residual degrees of freedom, and  $R^2$  alone says nothing about the evidence. 

**Exercise 2.13** (FREES exercise 3.2). With  $n = 100$ ,  $k = 3$ ,  $s = 50$  and

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 100 & 20 & 20 & 20 \\ 20 & 90 & 30 & 40 \\ 20 & 30 & 80 & 50 \\ 20 & 40 & 50 & 70 \end{pmatrix},$$

determine  $\text{SE}(b_3)$ , the estimated  $\text{cov}(b_2, b_3)$  and  $\text{cor}(b_2, b_3)$ , and the estimated variance of  $4b_2 + 3b_3$ .

**Solution.** By Theorem 2.17 the estimated covariance matrix is  $s^2(\mathbf{X}'\mathbf{X})^{-1}$ , so  $\text{SE}(b_j) = s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$  and  $\widehat{\text{cov}}(b_i, b_j) = s^2[(\mathbf{X}'\mathbf{X})^{-1}]_{ij}$ . The correlation cancels  $s^2$  entirely, so it can be read off  $(\mathbf{X}'\mathbf{X})^{-1}$  without knowing  $s$ . For a linear combination, Lemma 2.15 with  $\mathbf{c} = (0, 0, 4, 3)'$  gives  $\widehat{\text{var}}(\mathbf{c}'\boldsymbol{\beta}) = s^2\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}$ , which expands to  $s^2\{16(80) + 9(70) + 2(4)(3)(50)\}$ .

```
1 XtXi <- rbind(c(100, 20, 20, 20), c(20, 90, 30, 40),
2               c(20, 30, 80, 50),  c(20, 40, 50, 70))
3 sF <- 50; cF <- c(0, 0, 4, 3)
4 c(se_b3    = sF*sqrt(XtXi[4, 4]),
```

```

5 cov_b23 = sF^2*XtXi[3, 4],
6 cor_b23 = XtXi[3, 4]/sqrt(XtXi[3, 3]*XtXi[4, 4]),
7 var_lin = sF^2*drop(t(cF) %*% XtXi %*% cF),
8 se_lin  = sF*sqrt(drop(t(cF) %*% XtXi %*% cF)))

```

```

##          se_b3          cov_b23          cor_b23          var_lin
##      418.3300    125000.0000          0.6682    7775000.0000
##          se_lin
##      2788.3687

```

The correlation 0.668 is high enough that the two coefficients cannot be interpreted separately: the covariance term contributes 3,000,000 of the 7,775,000 total variance of  $4b_2 + 3b_3$ , so ignoring it would understate the standard error by a factor of  $\sqrt{7775/4775} = 1.28$ . ■

**Exercise 2.14** (FREES exercise 3.3). For the data

$$\mathbf{x}_1 = (-1, 2, 4, 6)', \quad \mathbf{x}_2 = (0, 0, 1, 1)', \quad \mathbf{y} = (0, 1, 5, 8)',$$

fit  $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$  by hand: write  $\mathbf{X}$ , compute  $\hat{\boldsymbol{\beta}}$ ,  $s$ ,  $\hat{y}_3$ ,  $\text{SE}(b_2)$  and  $t(b_1)$ . How many residual degrees of freedom are there, and what does that do to the test?

**Solution.**  $\mathbf{X}$  has the intercept column first, so it is  $4 \times 3$  and  $n - k - 1 = 4 - 3 = 1$ .

```
1 Xf <- cbind(1, x1 = c(-1, 2, 4, 6), x2 = c(0, 0, 1, 1))
2 yf <- c(0, 1, 5, 8)
3 Af <- solve(crossprod(Xf)); bf <- drop(Af %*% crossprod(Xf, yf))
4 ef <- yf - Xf %*% bf; sf <- sqrt(sum(ef^2)/(4 - 3))
5 round(Af, 5)
```

```
##           x1      x2
##      0.53846 -0.07692 -0.1538
## x1 -0.07692  0.15385 -0.6923
## x2 -0.15385 -0.69231  4.1154
```

```
1 c(b0 = bf[1], b1 = bf[2], b2 = bf[3], s = sf,
2   yhat3 = drop(Xf[3, ] %*% bf), se_b2 = sf*sqrt(Af[3, 3]),
3   t_b1 = bf[2]/(sf*sqrt(Af[2, 2])),
4   p_b1 = 2*pt(abs(bf[2]/(sf*sqrt(Af[2, 2]))), df = 1, lower.tail = FALSE))
```

```
##      b0    b1.x1    b2.x2      s    yhat3    se_b2 t_b1.x1
## 0.1538 0.6923 2.8846 1.3728 5.8077 2.7849 1.2857
## p_b1.x1
## 0.4208
```

By Theorem 2.5,  $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (0.1538, 0.6923, 2.8846)'$ ; by Definition 2.14,  $s = \sqrt{\text{SSE}/1} = 1.3728$ ; the third fitted value is  $0.1538 + 0.6923(4) + 2.8846 = 5.8077$ ; and by Theorem 2.17,  $\text{SE}(b_2) = s\sqrt{4.1154} = 2.785$  and  $t(b_1) = 0.6923/(1.3728\sqrt{0.15385}) = 1.286$ .

With one residual degree of freedom the test is nearly powerless:  $t_1$  has such heavy tails that the two-sided  $p$ -value at  $|t| = 1.286$  is 0.42, and the 5% critical value is 12.7. Four observations and three parameters leave almost nothing with which to estimate  $\sigma^2$ , so A4's requirement  $n > k + 1$  is met by the narrowest possible margin. ■

**Exercise 2.15.** Derive the variance of a fitted value at a new  $\mathbf{x}_0$  and the variance of the prediction error for a new response there, and hence the two intervals. Show  $\text{var}(\hat{Y}_i) = \sigma^2 h_{ii}$  at an observed point, and check that  $\text{var}(\hat{Y}_i) + \text{var}(e_i) = \sigma^2$ . Evaluate at `lstat` = 10, `age` = 60, `rm` = 6.5 and explain the ratio of the two widths.

**Solution.**  $\hat{Y}_0 = \mathbf{x}_0'\hat{\beta}$  is a linear map of  $\hat{\beta}$ , so Lemma 2.15 and Theorem 2.17 give  $\text{var}(\hat{Y}_0) = \sigma^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$ . A new response is  $Y_0 = \mathbf{x}_0'\beta + \varepsilon_0$  with  $\varepsilon_0$  independent of the sample, so  $\text{var}(Y_0 - \hat{Y}_0) = \sigma^2\{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0\}$ . Replacing  $\sigma$  by  $s$  and using Theorems 2.26 and 2.27 gives  $t_{n-k-1}$

intervals  $\hat{Y}_0 \pm t s \sqrt{\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$  for  $\mathbf{E}[Y_0]$  and  $\hat{Y}_0 \pm t s \sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$  for  $Y_0$ .

At an observed point  $\mathbf{x}_0 = \mathbf{x}_i$  the quadratic form is  $h_{ii}$ , so  $\text{var}(\hat{Y}_i) = \sigma^2 h_{ii}$  and, with Theorem 2.21,  $\text{var}(\hat{Y}_i) + \text{var}(e_i) = \sigma^2 h_{ii} + \sigma^2(1 - h_{ii}) = \sigma^2$ : leverage is the share of  $\sigma^2$  that goes into the fitted value, and  $1 - h_{ii}$  the share left in the residual. The two add because  $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$  and  $\mathbf{e} = \mathbf{M}\mathbf{Y}$  are uncorrelated,  $\mathbf{H}\mathbf{M} = \mathbf{0}$ .

```
1 x0 <- c(1, 10, 60, 6.5); nd0 <- data.frame(lstat = 10, age = 60, rm = 6.5)
2 lev0 <- drop(t(x0) %*% solve(XtX) %*% x0); tq <- qt(0.975, n - k - 1)
3 pr <- predict(fit, nd0, se.fit = TRUE)
4 c(se_fit_manual = sqrt(s2*lev0), se_fit_R = pr$se.fit,
5   CI_width = 2*tq*sqrt(s2*lev0), PI_width = 2*tq*sqrt(s2*(1 + lev0)),
6   ratio = sqrt((1 + lev0)/lev0))
```

```
## se_fit_manual      se_fit_R      CI_width      PI_width
##          0.2658          0.2658          1.0444          21.8023
##          ratio
##          20.8763
```

```
1 rbind(manual = c(pr$fit + c(-1, 1)*tq*sqrt(s2*lev0),
2                 pr$fit + c(-1, 1)*tq*sqrt(s2*(1 + lev0))),
```

```

3      R      = c(predict(fit, nd0, interval = "confidence")[2:3],
4                predict(fit, nd0, interval = "prediction")[2:3]))

```

```

##      [,1] [,2] [,3] [,4]
## manual 24.79 25.83 14.41 36.21
## R      24.79 25.83 14.41 36.21

```

The ratio is  $\sqrt{(1+h)/h}$ , which is 21 here. The confidence interval shrinks like  $n^{-1/2}$  and would close entirely with enough data; the prediction interval cannot fall below  $2ts$ , a new observation carrying its own  $\varepsilon_0$ . For  $n = 200$ ,  $k = 9$  and  $h_{ii} = 0.31$  with  $s = 12.4$ , the same formulas give  $\text{SE}(\hat{Y}_i) = 12.4\sqrt{0.31} = 6.90$  and  $\text{sd}(e_i) = 12.4\sqrt{0.69} = 10.30$ , and the point is flagged since 0.31 is far past the conventional  $2(k+1)/n = 0.10$ . ■

**Exercise 2.16.** Using Lemma 2.34, show  $\text{SSR} = \|(\mathbf{H} - \mathbf{H}_0)\mathbf{Y}\|^2$  and that the overall  $F$  compares mean squared lengths in two orthogonal subspaces of dimensions  $k$  and  $n - k - 1$  whose direct sum is  $\mathbf{1}^\perp$ . Deduce that  $F$  is unchanged if a constant is added to every  $Y_i$  or if  $\mathbf{Y}$  is multiplied by a non-zero constant.

**Solution.** The intercept makes  $\mathbf{H}_0\mathbf{Y} = \bar{Y}\mathbf{1}$  and  $\mathbf{H}\mathbf{H}_0 = \mathbf{H}_0$ , so  $\hat{\mathbf{Y}} - \bar{Y}\mathbf{1} = (\mathbf{H} - \mathbf{H}_0)\mathbf{Y}$  and SSR is its squared length; idempotency and symmetry give  $\|(\mathbf{H} - \mathbf{H}_0)\mathbf{Y}\|^2 = \mathbf{Y}'(\mathbf{H} - \mathbf{H}_0)\mathbf{Y}$ , and likewise  $\text{SSE} = \mathbf{Y}'\mathbf{M}\mathbf{Y}$ .

The two subspaces are  $\mathcal{C}(\mathbf{H} - \mathbf{H}_0)$ , of dimension  $\text{tr}(\mathbf{H} - \mathbf{H}_0) = k$ , and  $\mathcal{C}(\mathbf{M})$ , of dimension  $n - k - 1$ . They are orthogonal because  $(\mathbf{H} - \mathbf{H}_0)\mathbf{M} = \mathbf{0}$ , and  $(\mathbf{H} - \mathbf{H}_0) + \mathbf{M} = \mathbf{I} - \mathbf{H}_0$ , the projection onto  $\mathbf{1}^\perp$ , so they sum to it and the dimensions add to  $n - 1$ . Then

$$F = \frac{\|(\mathbf{H} - \mathbf{H}_0)\mathbf{Y}\|^2/k}{\|\mathbf{M}\mathbf{Y}\|^2/(n - k - 1)}.$$

Adding  $c\mathbf{1}$  changes neither:  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  gives  $(\mathbf{H} - \mathbf{H}_0)\mathbf{1} = \mathbf{0}$  and  $\mathbf{M}\mathbf{1} = \mathbf{0}$ . Multiplying  $\mathbf{Y}$  by  $a \neq 0$  multiplies both by  $a^2$ , which cancels. So  $F$ , and with it  $R^2$  by Corollary 2.36, is invariant to the origin and scale of the response: reporting in dollars or thousands of dollars changes every coefficient and no test statistic. It is not invariant to a non-affine transform such as a logarithm, which changes the subspaces the projections act on. ■

**Exercise 2.17.** The table after Definition 2.2 says normality is required for none of its first four rows. For each of  $\mathbf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ ,  $\text{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ ,  $\mathbf{E}[\text{MSE}] = \sigma^2$ , Gauss–Markov,  $\text{SST} = \text{SSR} + \text{SSE}$  and  $R^2 \leq 1$ , say which of A1–A5 the proof uses. Which results in the unit are false without A5?

**Solution.**  $\mathbf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$  uses A1, A2 for the mean and A4 for existence.  $\text{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$  adds A3: Lemma 2.15 is an identity about second moments.  $\mathbf{E}[\text{MSE}] = \sigma^2$  likewise, through Lemma 2.19, which needs only  $\mathbf{E}[\boldsymbol{\epsilon}] = \mathbf{0}$  and  $\text{var}(\boldsymbol{\epsilon}) = \sigma^2\mathbf{I}$ . Gauss–Markov uses A1–A4 and its proof is algebra on  $\mathbf{C}\mathbf{C}'$ .  $\text{SST} = \text{SSR} + \text{SSE}$  and  $0 \leq R^2 \leq 1$  use no probabilistic assumption at all: they hold for any  $\mathbf{Y}$  and any  $\mathbf{X}$  containing  $\mathbf{1}$ .

The results that fail without A5 are the distributional ones: Theorem 2.26, Theorem 2.27, Theorem 2.35 and Theorem 2.28. The third fails as an **approximation** too unless a central limit theorem applies, which is a condition on the design as well as the errors, specifically on whether one observation’s weight dominates. Note also that **uncorrelated** in A3 becomes **independent** only under A5, by Theorem 2.23; that is why Step 3 of Theorem 2.27 can conclude independence from a zero cross-covariance. ■

**Exercise 2.18.** Ridge regression,  $\hat{\beta}_\lambda = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}'\mathbf{Y}$  with  $\lambda > 0$ , is linear in  $\mathbf{Y}$  and can have smaller mean squared error than OLS. Which hypothesis of Theorem 2.18 does it violate?

**Solution.** Unbiasedness, not linearity. With  $\mathbf{A}_\lambda = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}'$ ,  $\hat{\beta}_\lambda = \mathbf{A}_\lambda\mathbf{Y}$  is linear, but

$$\mathbf{E}[\hat{\beta}_\lambda] = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}'\mathbf{X}\beta \neq \beta \quad (\lambda > 0),$$

so it lies outside the class Theorem 2.18 optimises over. The theorem compares variances within the unbiased class; mean squared error is bias<sup>2</sup> plus variance, and ridge accepts bias to reduce variance. Shrinking  $\hat{\beta}$  toward zero by any small enough factor always helps, the variance falling at first order in the shrinkage while the squared bias grows only at second order. ■

**Exercise 2.19.** Take  $k = 1$  with no intercept,  $\hat{\beta} = \mathbf{x}'\mathbf{Y}/S$  with  $S = \mathbf{x}'\mathbf{x}$ , and consider  $c\hat{\beta}$  for  $c \in [0, 1]$ . Compute its mean squared error, find the minimising  $c^*$ , show  $c^* < 1$  always, and evaluate at  $S = 100$ ,  $\sigma^2 = 4$ ,  $\beta = 1$ . Why can  $c^*$  not be used in practice?

**Solution.**  $E[\hat{\beta}] = \beta$  and  $\text{var}(\hat{\beta}) = \sigma^2/S$ , so  $c\hat{\beta}$  has bias  $(c - 1)\beta$  and variance  $c^2\sigma^2/S$ :

$$\text{MSE}(c) = c^2 \frac{\sigma^2}{S} + (1 - c)^2 \beta^2.$$

This is a convex quadratic; setting  $2c\sigma^2/S - 2(1 - c)\beta^2 = 0$ ,

$$c^* = \frac{\beta^2}{\beta^2 + \sigma^2/S} \in (0, 1),$$

strictly less than 1 for every  $\sigma^2 > 0$  and finite  $S$ . Substituting,

$$\text{MSE}(c^*) = \frac{\beta^2 \sigma^2 / S}{\beta^2 + \sigma^2 / S} = c^* \cdot \frac{\sigma^2}{S} = c^* \text{MSE}(1),$$

so the best shrinkage improves on OLS by exactly the factor  $c^*$ . At  $S = 100$ ,  $\sigma^2 = 4$ ,  $\beta = 1$ :  $\sigma^2/S = 0.04$ ,  $c^* = 0.9615$ ,  $\text{MSE}(\text{OLS}) = 0.04$  and  $\text{MSE}(c^*\hat{\beta}) = 0.03846$ , an improvement of 3.8%.  $c^*$  depends on  $\beta$ , the unknown being estimated. What can be done is to choose the amount of shrinkage by an estimate of prediction error. The gain is small here because  $\beta^2$  is large relative to  $\sigma^2/S$ ; the improvement factor  $\{1 + \sigma^2/(S\beta^2)\}^{-1}$  is a statement about the signal-to-noise ratio. ■

**Exercise 2.20.** The true model is  $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$  but only  $\mathbf{X}_1$  is fitted. Show  $\mathbf{E}[\hat{\boldsymbol{\beta}}_1] = \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2$ , state when the omission causes no bias, and give the sign of the bias on a single slope when the omitted coefficient and the correlation are both positive.

**Solution.** Substituting the true  $\mathbf{Y}$  into  $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{Y}$ ,

$$\hat{\boldsymbol{\beta}}_1 = \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\boldsymbol{\varepsilon},$$

and the last term has expectation zero by [A2](#). The extra term is the coefficient matrix from regressing  $\mathbf{X}_2$  on  $\mathbf{X}_1$ , times  $\boldsymbol{\beta}_2$ ; it vanishes iff  $\boldsymbol{\beta}_2 = \mathbf{0}$  or  $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$ , which with an intercept means the two are uncorrelated in the sample.

With one included and one omitted variable the term is  $\delta\beta_2$ , and  $\delta$  has the sign of  $\text{cor}(x_1, x_2)$  by [Corollary 2.39](#). Both positive means the fitted slope is biased **upward**: it carries its own effect plus the part of  $x_2$ 's effect travelling with  $x_1$ . This is why a coefficient can change sign when a control is added. 

**Exercise 2.21.** Show that centering every non-intercept column leaves the slopes, residuals, SSE and  $R^2$  unchanged and sets the intercept to  $\bar{Y}$ ; and that multiplying one column by  $c \neq 0$  divides its coefficient and standard error by  $c$ , leaving  $t$ , SSE and  $R^2$  unchanged. Verify on the fit of §8.1.

**Solution.** Both follow from Theorem 2.6: the fit depends on  $\mathbf{X}$  only through  $\mathcal{C}(\mathbf{X})$ . Centering replaces column  $j$  by  $\mathbf{x}_j - \bar{x}_j \mathbf{1}$ , a combination of  $\mathbf{x}_j$  and the intercept column already present; scaling replaces  $\mathbf{x}_j$  by  $c\mathbf{x}_j$ . In both cases  $\mathbf{X}_{\text{new}} = \mathbf{X}\mathbf{A}$  with  $\mathbf{A}$  invertible, so  $\mathcal{C}(\mathbf{X}_{\text{new}}) = \mathcal{C}(\mathbf{X})$  and  $\hat{\mathbf{Y}}$ ,  $\mathbf{e}$ , SSE, SST,  $R^2$  are identical. The coefficients transform as  $\hat{\boldsymbol{\beta}}_{\text{new}} = \mathbf{A}^{-1}\hat{\boldsymbol{\beta}}$  and their covariance as  $\mathbf{A}^{-1}\{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}\}\mathbf{A}^{-'}$ , so for diagonal  $\mathbf{A}$  with  $j$ -th entry  $c$  both the coefficient and its standard error divide by  $c$  and  $t$  is unchanged. Under centering  $\mathbf{A}^{-1}$  leaves the slopes and sends the intercept to  $\bar{Y} - \sum_j \hat{\beta}_j \bar{x}_j + \sum_j \hat{\beta}_j \bar{x}_j = \bar{Y}$ .

```

1 Bc <- transform(Boston, lstat = lstat - mean(lstat),
2                   age = age - mean(age), rm = rm - mean(rm))
3 mc <- lm(medv ~ lstat + age + rm, data = Bc)
4 Bs <- transform(Boston, lstat = 10*lstat)
5 ms <- lm(medv ~ lstat + age + rm, data = Bs)
6 c(slopes_moved = max(abs(coef(mc)[-1] - coef(fit)[-1])),

```

```

7 intercept_new = coef(mc)[[1]], ybar = mean(y),
8 coef_ratio   = coef(fit)[["lstat"]]/coef(ms)[["lstat"]],
9 t_moved      = coef(summary(ms))["lstat", 3] -
10               coef(summary(fit))["lstat", 3],
11 R2_moved     = summary(ms)$r.squared - summary(fit)$r.squared

```

```

## slopes_moved intercept_new      ybar      coef_ratio
##      1.776e-15      2.253e+01      2.253e+01      1.000e+01
##           t_moved      R2_moved
##      -3.553e-15      0.000e+00

```

A coefficient has units and its size means nothing until they are stated, while  $t$ ,  $R^2$  and SSE are unit-free. Standardising the predictors is therefore a presentational choice for least squares, and, since it changes  $\|\beta\|$ , is not one for a penalised method. 

**Exercise 2.22.** Simulate  $y_i = 1 + 2x_i + \varepsilon_i$  at  $n = 20$  and record the coverage of the nominal 95%  $t$  interval for the slope, for two designs and two error laws:  $x$  equally spaced on  $[-1, 1]$  against  $x = (0, \dots, 0, 10)$ , and standard normal errors against centred exponential ones. Predict the pattern first.

**Solution.**  $\hat{\beta}_1 - \beta_1 = \sum_i w_i \varepsilon_i$  with  $w_i = (x_i - \bar{x})/S_{xx}$ , a weighted sum of the errors. Under A5 it is exactly normal whatever the weights, so the interval is exact. Without A5 it is approximately normal only if no single weight dominates, which is a condition on the design. The balanced design spreads the weights; the second concentrates almost all of the weight on one observation, whose leverage is 1 (Exercise 2.6), so  $\hat{\beta}_1$  inherits that error's law, skew and all.

```

1 cover <- function(xx, gen, R = 8000, seed = 9) {
2   set.seed(seed); nn <- length(xx); b1 <- 2
3   ok <- lo <- hi <- logical(R)
4   for (r in seq_len(R)) {
5     yy <- 1 + b1*xx + gen(nn)
6     ci <- confint(lm(yy ~ xx))["xx", ]
7     ok[r] <- ci[1] <= b1 && b1 <= ci[2]
8     lo[r] <- b1 > ci[2]; hi[r] <- b1 < ci[1]
9   }
10  c(cover = mean(ok), miss_below = mean(lo), miss_above = mean(hi))
11 }
12 bal <- seq(-1, 1, length.out = 20); lev <- c(rep(0, 19), 10)
13 rbind(normal_balanced = cover(bal, function(m) rnorm(m)),

```

```

14 normal_leverage = cover(lev, function(m) rnorm(m)),
15 expon_balanced   = cover(bal, function(m) rexp(m) - 1),
16 expon_leverage   = cover(lev, function(m) rexp(m) - 1))

```

```

##               cover miss_below miss_above
## normal_balanced 0.9513      0.02337    0.02538
## normal_leverage 0.9477      0.02738    0.02488
## expon_balanced  0.9489      0.02388    0.02725
## expon_leverage  0.9375      0.00000    0.06250

```

```

1 c(max_lev_balanced = max(hatvalues(lm(rnorm(20) ~ bal))),
2   max_lev_unbalanced = max(hatvalues(lm(rnorm(20) ~ lev))))

```

```

## max_lev_balanced max_lev_unbalanced
##               0.1857                1.0000

```

Three of the four rows are at nominal coverage: with a balanced design the  $t$  interval survives badly skewed errors, the weighted sum being close to normal. The fourth loses a point of coverage but misses **entirely on one side**, so the interval is not merely short but systematically displaced. A symmetric error law would not do this at any leverage, and a balanced design would not do it under any error law. The quantity to inspect before dropping [A5](#) is the maximum leverage, not a

histogram of the residuals. 

**Exercise 2.23.** On the fit of §8.1: (a) show  $\hat{\mathbf{Y}}'\mathbf{e} = 0$  holds to machine precision but that `cor( $\hat{\mathbf{Y}}$ ,  $\mathbf{e}$ )` prints as a much smaller number, and explain; (b) construct a fit whose  $\bar{R}^2$  is negative and say what that means.

**Solution.**

```
1 c(inner_product = sum(fitted(fit)*e), correlation = cor(fitted(fit), e),
2   scale = sqrt(sum(fitted(fit)^2))*sqrt(sum(e^2)),
3   ratio = sum(fitted(fit)*e)/(sqrt(sum(fitted(fit)^2))*sqrt(sum(e^2))))
```

```
## inner_product    correlation          scale          ratio
##      2.248e-12      7.153e-17      6.620e+04      3.396e-17
```

(a) Both are zero in exact arithmetic and differ only in scale. Since  $\bar{e} = 0$  the correlation has the same numerator, divided by  $\|\hat{\mathbf{Y}}\| \|\mathbf{e}\| = 533 \times 124 = 6.6 \times 10^4$ . So the raw inner product carries that factor times the machine epsilon and prints at  $10^{-12}$ , while the correlation prints at  $10^{-17}$ . A numerical check must be read against the scale of the quantities forming it, which is why the checks of §8 print differences of  $O(1)$  quantities.

```

1  set.seed(21)
2  dsub <- Boston[1:30, ]
3  rep500 <- replicate(500, {
4    Nz <- matrix(rnorm(30*10), 30, 10)
5    sm <- summary(lm(dsub$medv ~ Nz))
6    c(R2 = sm$r.squared, adjR2 = sm$adj.r.squared)
7  })
8  c(mean_R2 = mean(rep500["R2", ]), threshold = 10/29,
9    mean_adjR2 = mean(rep500["adjR2", ]),
10   frac_adjR2_negative = mean(rep500["adjR2", ] < 0))

```

```

##              mean_R2              threshold              mean_adjR2
##              0.350837              0.344828              0.009173
## frac_adjR2_negative
##              0.512000

```

(b)  $\bar{R}^2 = 1 - \frac{n-1}{n-k-1}(1 - R^2)$  is negative iff  $R^2 < k/(n-1)$ . Ten columns of pure noise on  $n = 30$  give  $R^2$  averaging  $k/(n-1) = 0.345$  by chance alone, so  $\bar{R}^2$  straddles zero and is negative in half the draws. A negative  $\bar{R}^2$  means the fitted model has a larger residual mean square than the intercept-only model, that is, it predicts worse than  $\bar{Y}$  once the parameter count is charged. It

cannot happen to  $R^2$  (Proposition 2.32). ■

**Exercise 2.24** (FREES exercise 2.9). Suppose  $x_i$  takes only the values 0 and 1:  $n_1$  observations have  $x = 0$  with mean response  $\bar{y}_1$ , and  $n_2 = n - n_1$  have  $x = 1$  with mean  $\bar{y}_2$ . Show  $b_1 = \bar{y}_2 - \bar{y}_1$  and  $b_0 = \bar{y}_1$ , and give  $\text{SE}(b_1)$  in terms of  $n_1$ ,  $n_2$  and  $s$ . What familiar test is  $t(b_1)$ ?

**Solution.** Here  $\bar{x} = n_2/n$  and  $\sum_i (x_i - \bar{x})^2 = n_2(1 - \bar{x})^2 + n_1\bar{x}^2 = n_1n_2/n$ , since  $x_i^2 = x_i$ . Also  $\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i - n\bar{x}\bar{y} = n_2\bar{y}_2 - \frac{n_2}{n}(n_1\bar{y}_1 + n_2\bar{y}_2) = \frac{n_1n_2}{n}(\bar{y}_2 - \bar{y}_1)$ . By Corollary 2.39 the ratio is

$$b_1 = \bar{y}_2 - \bar{y}_1, \quad b_0 = \bar{y} - b_1\bar{x} = \bar{y}_1.$$

Fitted values are the two group means, so SSE is the within-group sum of squares and  $s^2$  is the pooled variance on  $n - 2$  degrees of freedom. By Corollary 2.40,

$$\text{SE}(b_1) = \frac{s}{\sqrt{\sum_i (x_i - \bar{x})^2}} = s\sqrt{\frac{n}{n_1n_2}} = s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}},$$

the last step because  $n/(n_1n_2) = 1/n_1 + 1/n_2$ . So  $t(b_1) = (\bar{y}_2 - \bar{y}_1)/\{s\sqrt{1/n_1 + 1/n_2}\}$  is exactly

the pooled two-sample  $t$  statistic: regression on a single indicator **is** the two-sample comparison, with A3 playing the role of the equal-variance assumption and  $n - 2$  the pooled degrees of freedom.

**Exercise 2.25** (FREES exercise 2.6). Let  $b_{y|x}$  be the slope from regressing  $y$  on  $x$  and  $b_{x|y}$  that from regressing  $x$  on  $y$ , both with intercepts. Show  $|r| = \sqrt{b_{y|x} b_{x|y}}$ , that the two lines coincide only when  $|r| = 1$ , and deduce the regression-to-the-mean statement.

**Solution.** By Corollary 2.39,  $b_{y|x} = r s_y/s_x$  and, exchanging roles,  $b_{x|y} = r s_x/s_y$ . Their product is  $r^2$ , so  $|r| = \sqrt{b_{y|x} b_{x|y}}$ : the correlation is the geometric mean of the two slopes, and the two have the same sign, that of  $r$ .

Drawn on the same axes, the second line has slope  $1/b_{x|y} = s_y/(r s_x)$  against the first's  $r s_y/s_x$ ; the ratio is  $r^2$ , so they coincide iff  $r^2 = 1$ , that is iff the points are collinear (Proposition 2.38). Otherwise the regression of  $y$  on  $x$  is always the flatter of the two, and which line one draws depends on which variable is being predicted.

Standardising both variables gives  $s_x = s_y = 1$ , hence  $b_{y|x} = r$  and  $b_0 = 0$ : the fitted line is  $\hat{y} = rx$  in standard units, so a point one standard deviation above average in  $x$  is predicted only  $r \leq 1$

standard deviations above average in  $y$ . Nothing is changing over time and no force is acting: a conditional mean is less extreme than the conditioning value whenever  $|r| < 1$ . Selecting the extreme cases of one variable and measuring them on another therefore always shows movement toward the average. ■

**Exercise 2.26** (ISLR §3.7, exercise 11). Generate  $x_i$  standard normal and  $y_i = 2x_i + \varepsilon_i$  with  $n = 100$ . (a) Regress  $y$  on  $x$  without an intercept and then  $x$  on  $y$  without an intercept, and compare the two  $t$  statistics. (b) Show algebraically that for regression through the origin

$$t = \frac{\sqrt{n-1} \sum_i x_i y_i}{\sqrt{(\sum_i x_i^2)(\sum_i y_i^2) - (\sum_i x_i y_i)^2}},$$

and use it to explain (a). (c) Check that the same symmetry holds with an intercept.

**Solution.** (b) With  $\hat{\beta} = \sum_i x_i y_i / \sum_i x_i^2$  and, for a model with one parameter and no intercept,

$$\text{SE}(\hat{\beta}) = \sqrt{\sum_i (y_i - x_i \hat{\beta})^2 / \{(n-1) \sum_i x_i^2\}},$$

$$t = \frac{\hat{\beta}}{\text{SE}(\hat{\beta})} = \frac{\sum_i x_i y_i}{\sum_i x_i^2} \cdot \frac{\sqrt{n-1} \sqrt{\sum_i x_i^2}}{\sqrt{\sum_i (y_i - x_i \hat{\beta})^2}}.$$

Expanding the residual sum of squares,

$$\sum_i (y_i - x_i \hat{\beta})^2 = \sum_i y_i^2 - 2\hat{\beta} \sum_i x_i y_i + \hat{\beta}^2 \sum_i x_i^2 = \sum_i y_i^2 - \frac{(\sum_i x_i y_i)^2}{\sum_i x_i^2},$$

so multiplying numerator and denominator by  $\sqrt{\sum_i x_i^2}$  gives the display. The expression is symmetric in  $x$  and  $y$ , so exchanging their roles cannot change it: the  $t$  statistic for  $H_0: \beta = 0$  is the same in both regressions, even though the two coefficients differ.

```
1 set.seed(1); xa <- rnorm(100); ya <- 2*xa + rnorm(100)
2 tf <- sqrt(99)*sum(xa*ya) /
3     sqrt(sum(xa^2)*sum(ya^2) - sum(xa*ya)^2)
4 c(t_y_on_x = coef(summary(lm(ya ~ xa + 0)))[1, 3],
```

```

5 t_x_on_y = coef(summary(lm(xa ~ ya + 0)))[1, 3],
6 closed_form = tf,
7 coef_y_on_x = coef(lm(ya ~ xa + 0))[[1]],
8 coef_x_on_y = coef(lm(xa ~ ya + 0))[[1]],
9 with_intercept_y_on_x = coef(summary(lm(ya ~ xa)))[2, 3],
10 with_intercept_x_on_y = coef(summary(lm(xa ~ ya)))[2, 3])

```

```

##           t_y_on_x           t_x_on_y
##           18.7259           18.7259
##           closed_form       coef_y_on_x
##           18.7259           1.9939
##           coef_x_on_y with_intercept_y_on_x
##           0.3911           18.5556
## with_intercept_x_on_y
##           18.5556

```

(a) and (c) The two  $t$  statistics agree exactly, with and without an intercept, while the coefficients 1.994 and 0.391 do not. The symmetry with an intercept is Exercise 2.25:  $t = b_{y|x}/\text{SE}(b_{y|x})$  reduces to  $r\sqrt{n-2}/\sqrt{1-r^2}$ , a function of  $r$  alone, and  $r$  is symmetric. Testing  $\beta_1 = 0$  is testing  $r = 0$ , a statement about the pair and not about a direction. ■

**Exercise 2.27.** A regression on  $n = 50$  observations has  $s = 3792$ ,  $s_x = 11,098$  and  $b_1 = 0.647$ . Compute  $\text{SE}(b_1)$ , test  $H_0: \beta_1 = 0$ , and give a 95% confidence interval. If  $R^2 = 0.785$ , what is  $r$ , and what is the overall  $F$ ?

**Solution.**  $\text{SE}(b_1) = 3792/(11,098\sqrt{49}) = 0.0488$ , so  $t = 0.647/0.0488 = 13.3$  on  $n - 2 = 48$  degrees of freedom. Since  $t_{0.975,48} = 2.011$ , reject  $H_0$ ; the interval is  $0.647 \pm 2.011(0.0488) = (0.549, 0.745)$ .

By Corollary 2.41,  $r = \pm\sqrt{0.785} = 0.886$ , positive because  $b_1 > 0$ . With  $k = 1$  the overall  $F$  is  $t^2 = 177$ , equivalently  $F = R^2/\{(1 - R^2)/(n - 2)\} = 0.785/(0.215/48) = 175$ , the difference being rounding. ■

**Exercise 2.28** (SOA sample question 23). A cafeteria sells coffee at 1.00 for 12 ounces, 1.20 for 16 and 1.40 for 20, and announces that any size will be priced at the value predicted by the least-squares regression of price on size. Toby and Karen each buy a 24-ounce coffee. Calculate what they save by buying one 48-ounce coffee instead. What are  $R^2$ , SSE and  $s$  here, and what does that do to  $t(b_1)$ ?

**Solution.** The three points are collinear, so by Proposition 2.38 the least-squares line passes through all three and no arithmetic is needed: the slope is  $(1.40 - 1.00)/(20 - 12) = 0.05$  per ounce and the intercept is  $1.00 - 0.05(12) = 0.40$ . The price of  $s$  ounces is  $0.40 + 0.05s$ : a fixed 0.40 per cup plus 5 cents an ounce.

Two 24-ounce cups cost  $2(0.40 + 1.20) = 3.20$ ; one 48-ounce cup costs  $0.40 + 2.40 = 2.80$ . The saving is 0.40, exactly the one cup charge avoided, and would be zero if the regression were forced through the origin.

Collinearity gives  $\text{SSE} = 0$ ,  $R^2 = r^2 = 1$  by Corollary 2.41, and  $s = \sqrt{0/(n-2)} = 0$ . The statistic  $t(b_1) = 0.05/0$  is undefined. This is the correct reading of A3 and A5 with  $\sigma^2$  estimated as zero: three points and two parameters leave one residual degree of freedom, spent on a perfect fit, so the model carries no information about  $\sigma^2$ . A regression can be perfectly determined and completely untestable at once. ■